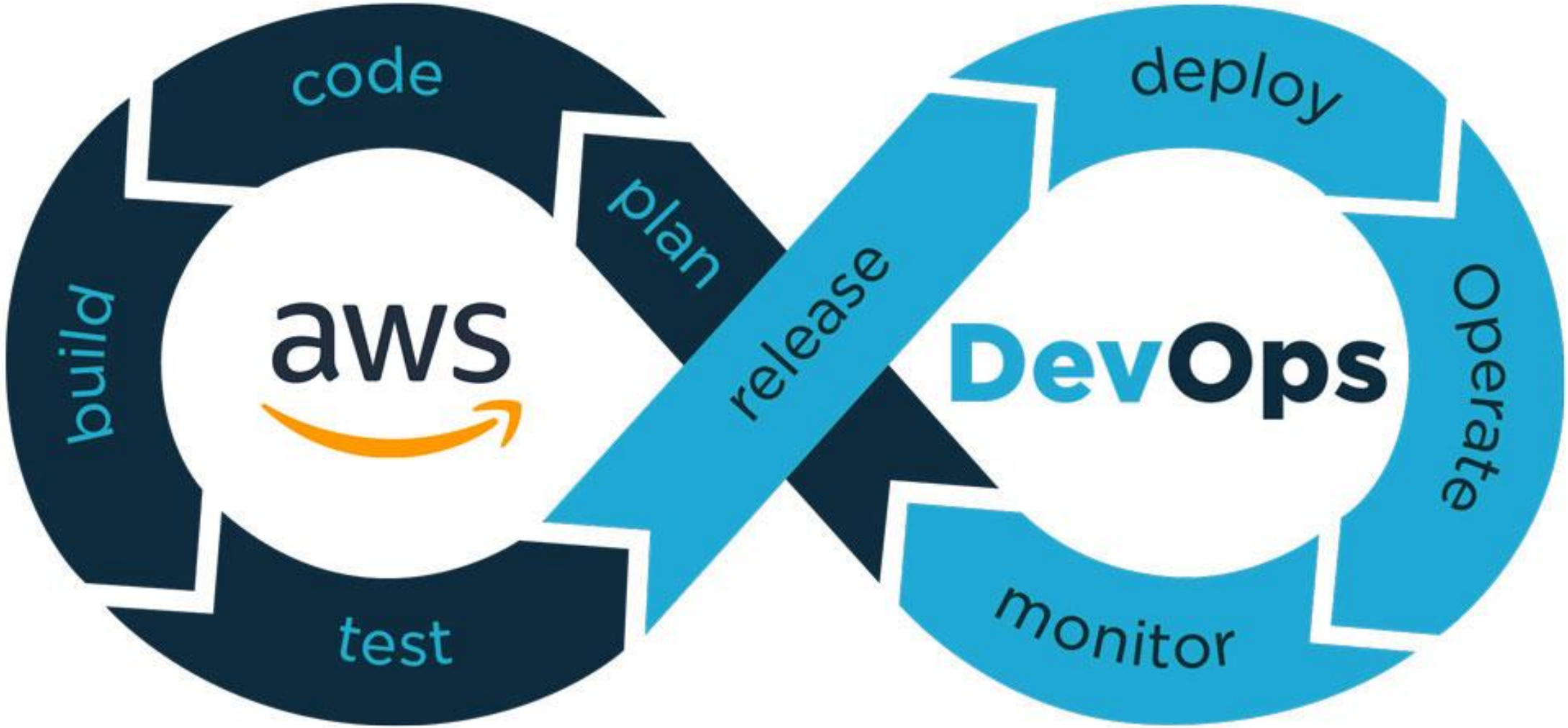

code
plan
build
test
aws
release
deploy
operate
monitor
DevOps

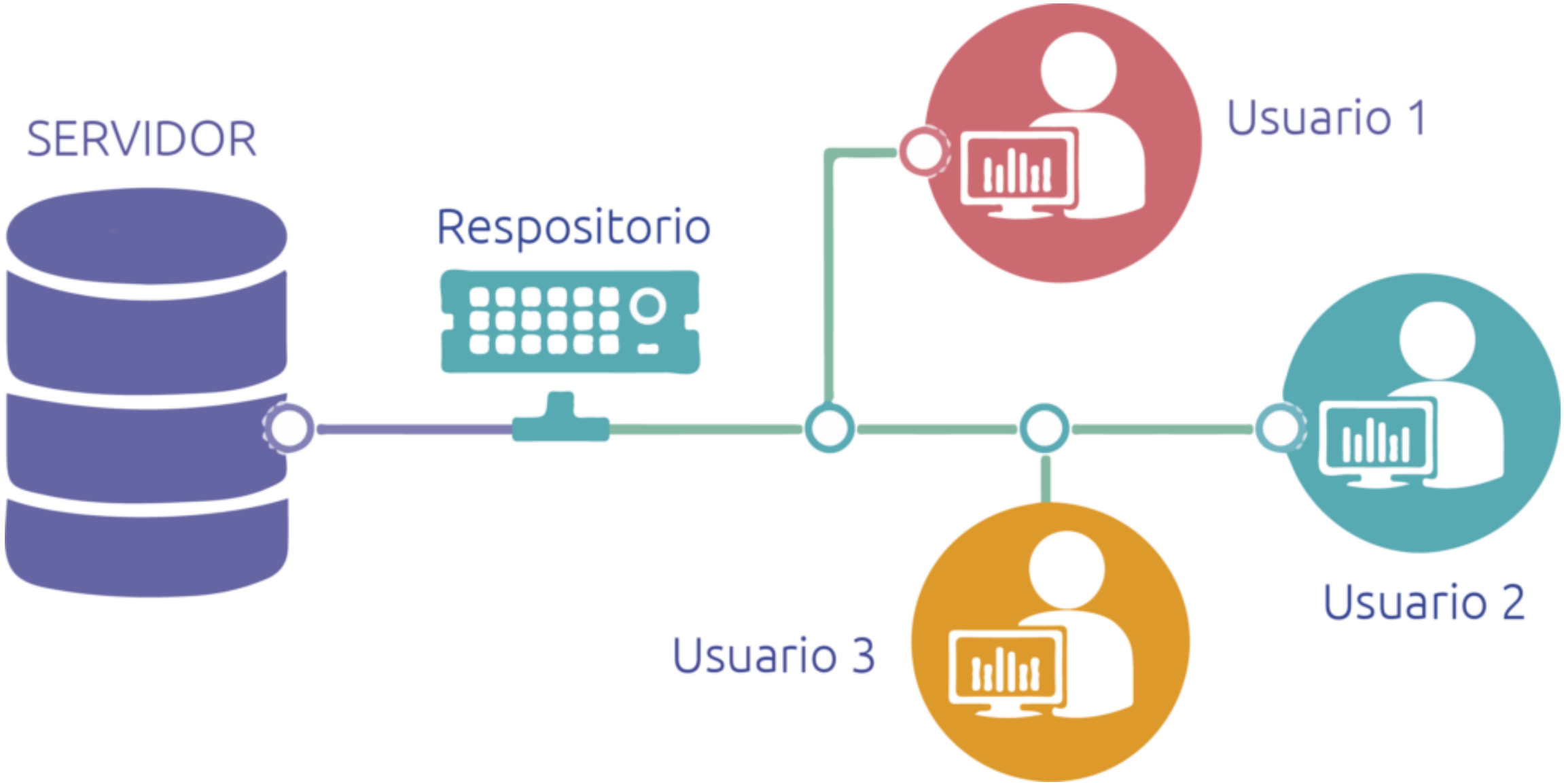
- **Version Control Systems**
  - Advanced Git concepts
  - Common branching strategies and when is better to use each one depending on the project situation and team configuration
  - Git best practices
    - Merging versus Rebasing - Advantages and disadvantages
    - Keeping a clean history
      - Stashing
      - Commit comments
    - Semantic versioning

- **Continuous Integration (CI)**
  - Principles of continuous integration
  - Creating a build pipeline for a dotnet solution using yaml configuration files tracked in the repository
  - Unit test integration
  - Pipeline triggers

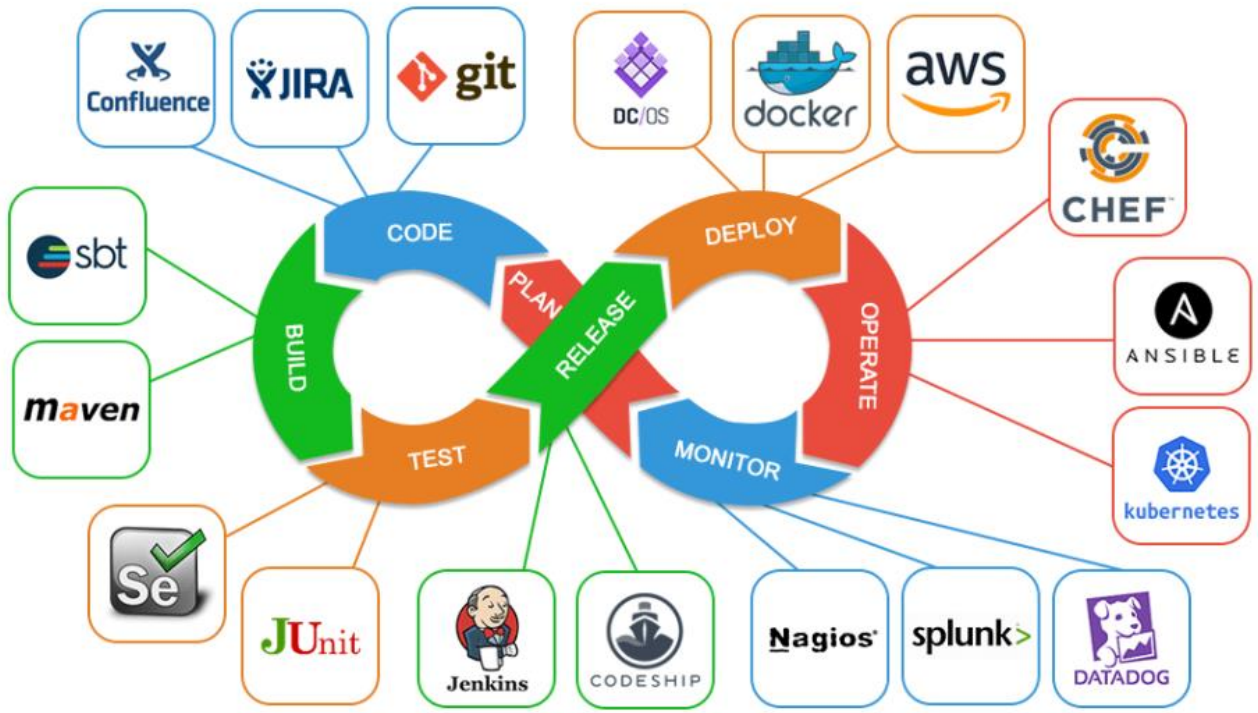
- **AWS Fundamentals**
  - Introduction to AWS (not detailed, most of the students will have this knowledge)
  - Basic AWS for deploying a dotnet solution
    - EC2 instances
    - S3
    - Lambdas
    - AWS Secret Manager
    - Cloud Watch

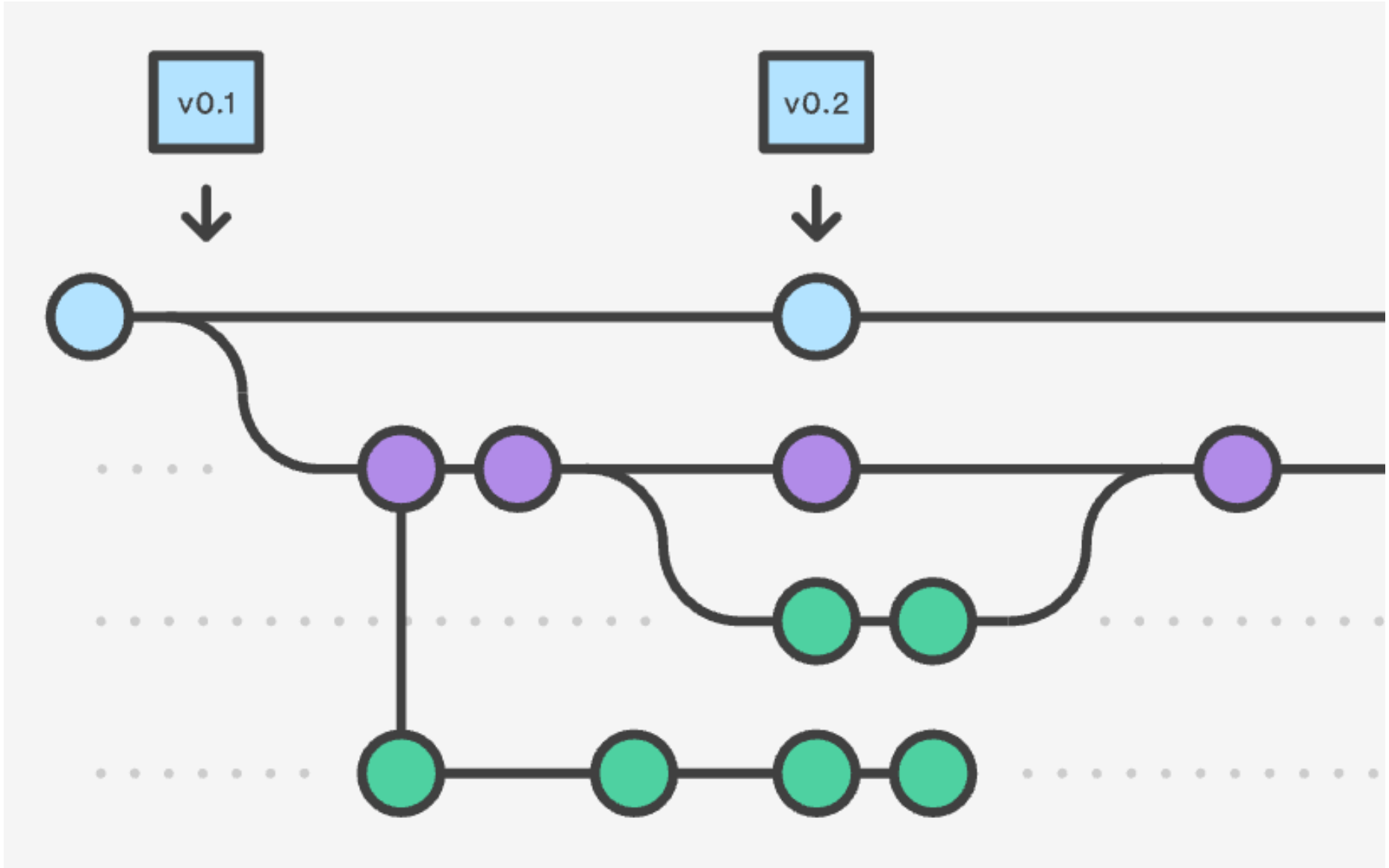
- **Continuous Delivery (CD)**
  - Principles of continuous delivery and deployment
  - Creating a Release pipeline using Jenkins or Github Actions
    - Using the artifacts generated from the build pipeline
    - Deploy into different environment
    - Approvals and gatekeepers
    - HealthChecks
    - Blue/Green deployments
    - Canary deployments
- **Introduction of Infrastructure as Code (IaC)**
  - Basics of Infrastructure as Code
  - Introduction to Terraform

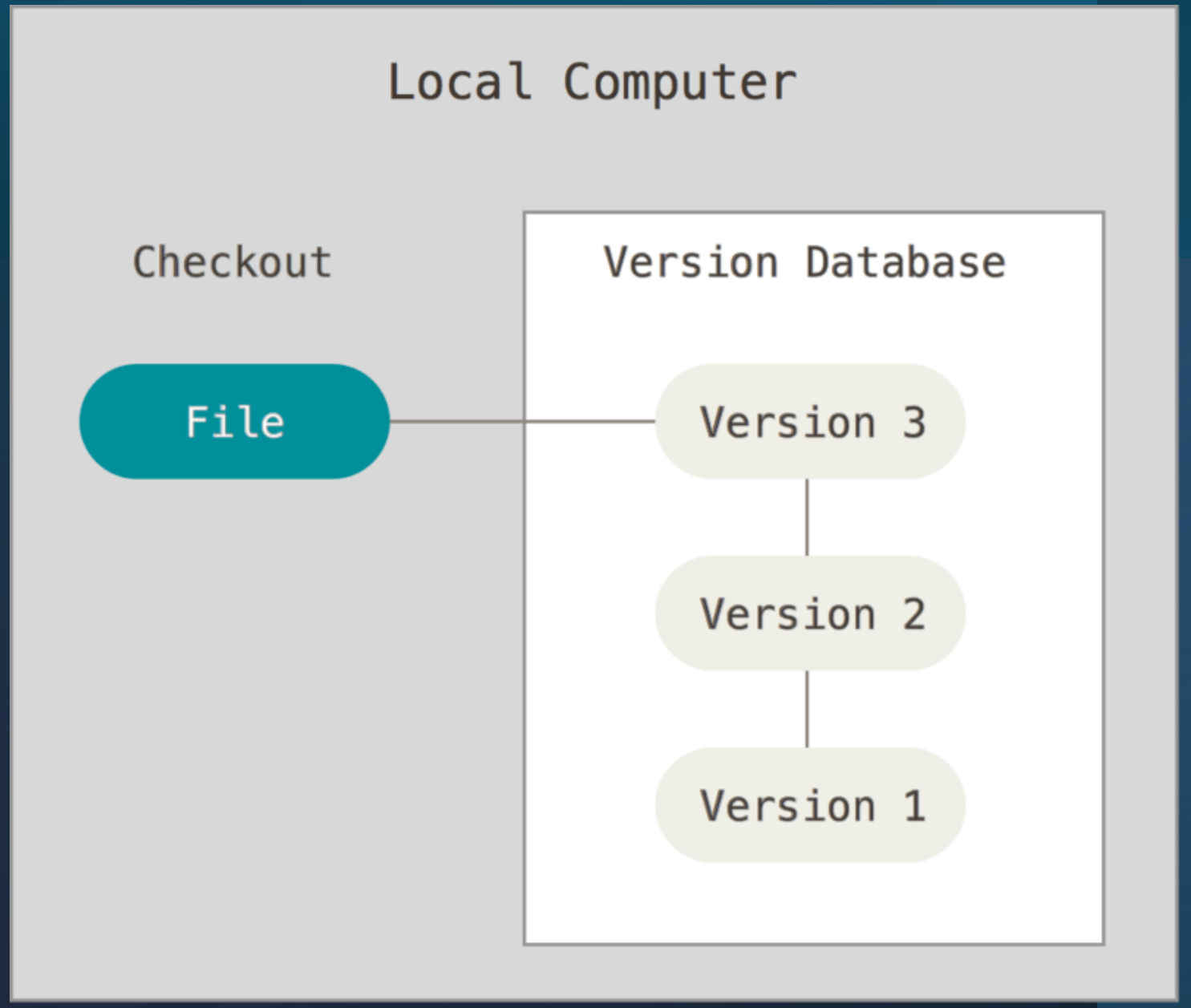
INTRODUCE YOURSELF

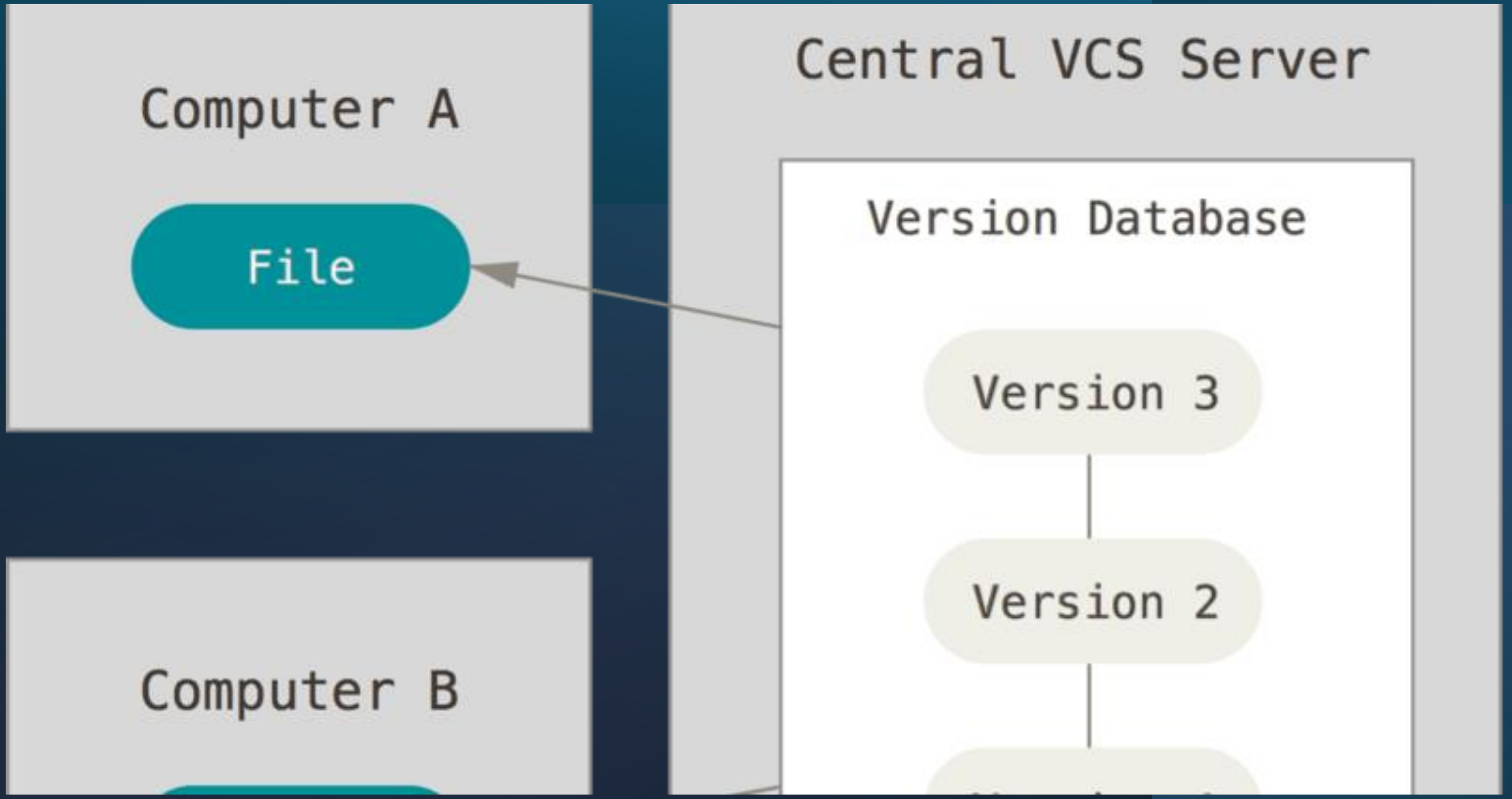

SERVIDOR
Respositorio
Usuario 1
Usuario 2
Usuario 3

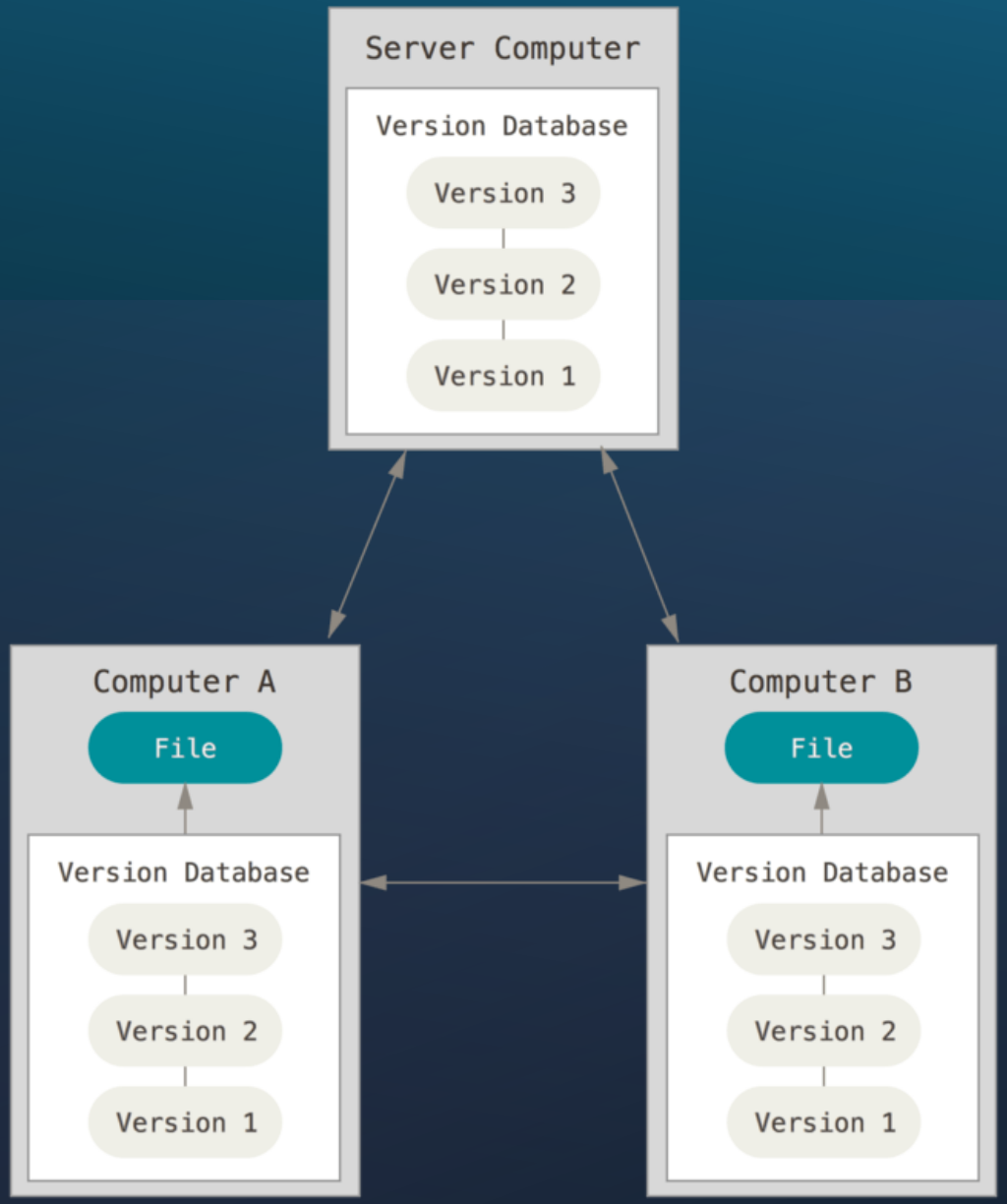
DEVOPS


Confluence
JIRA
git
DC/OS
docker
aws
CHEF
sbt
CODE
PLAN
DEPLOY
ANSIBLE
BUILD
RELEASE
OPERATE
maven
TEST
MONITOR
kubernetes
Se
JUnit
Jenkins
CODESHIP
Nagios
splunk>
DATADOG


v0.1
v0.2

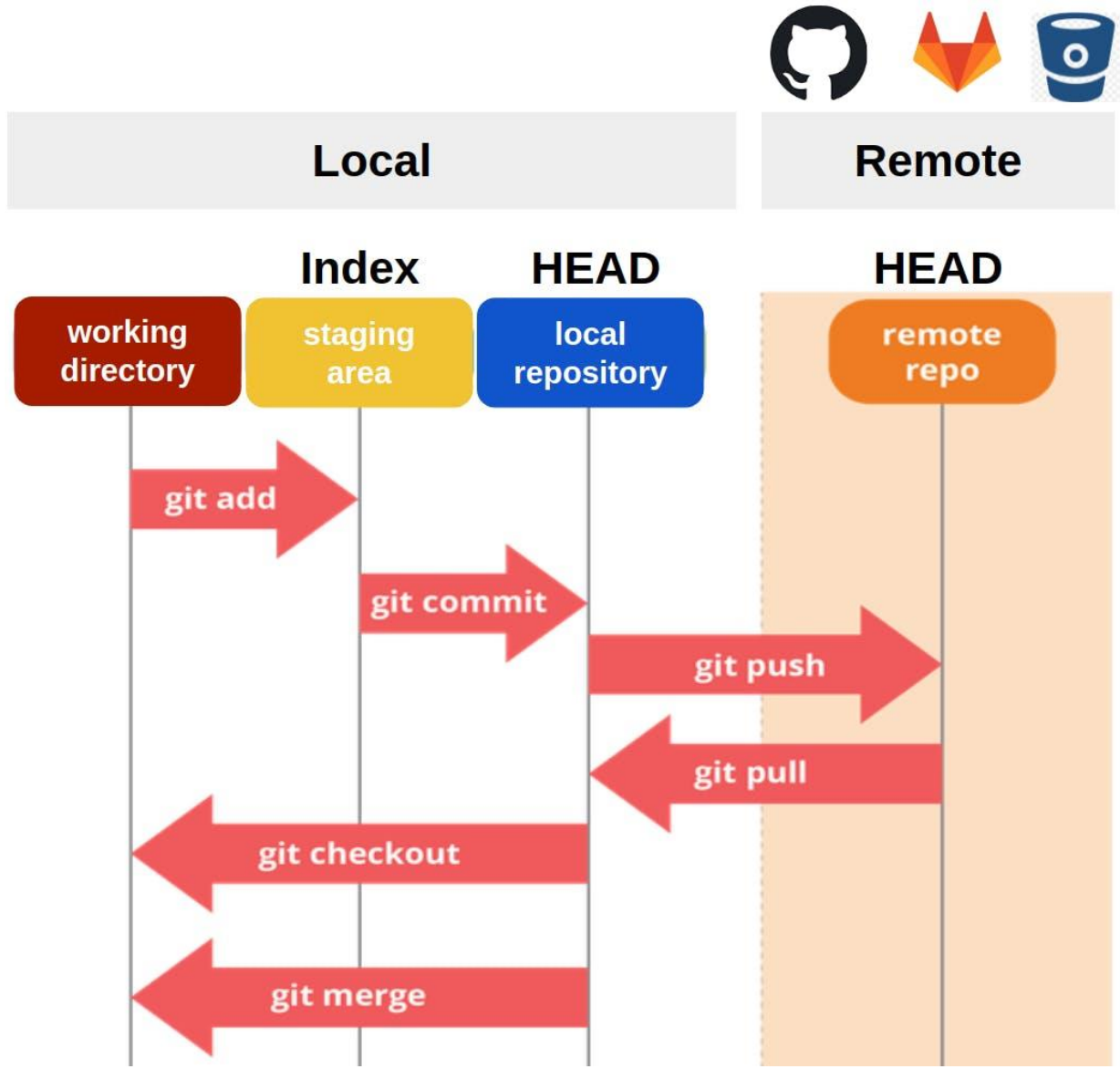

Local Computer
Checkout
File
Version Database
Version 3
Version 2
Version 1

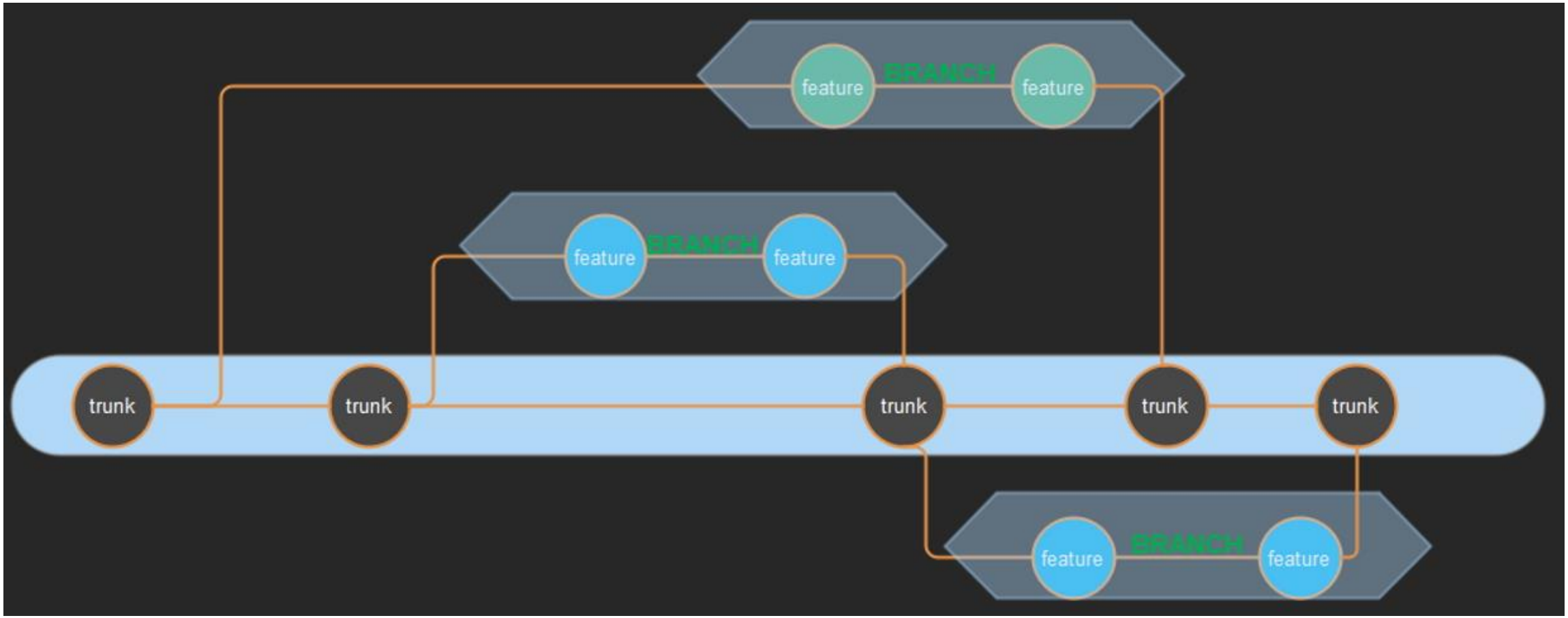

Computer A
File
Computer B
Central VCS Server
Version Database
Version 3
Version 2

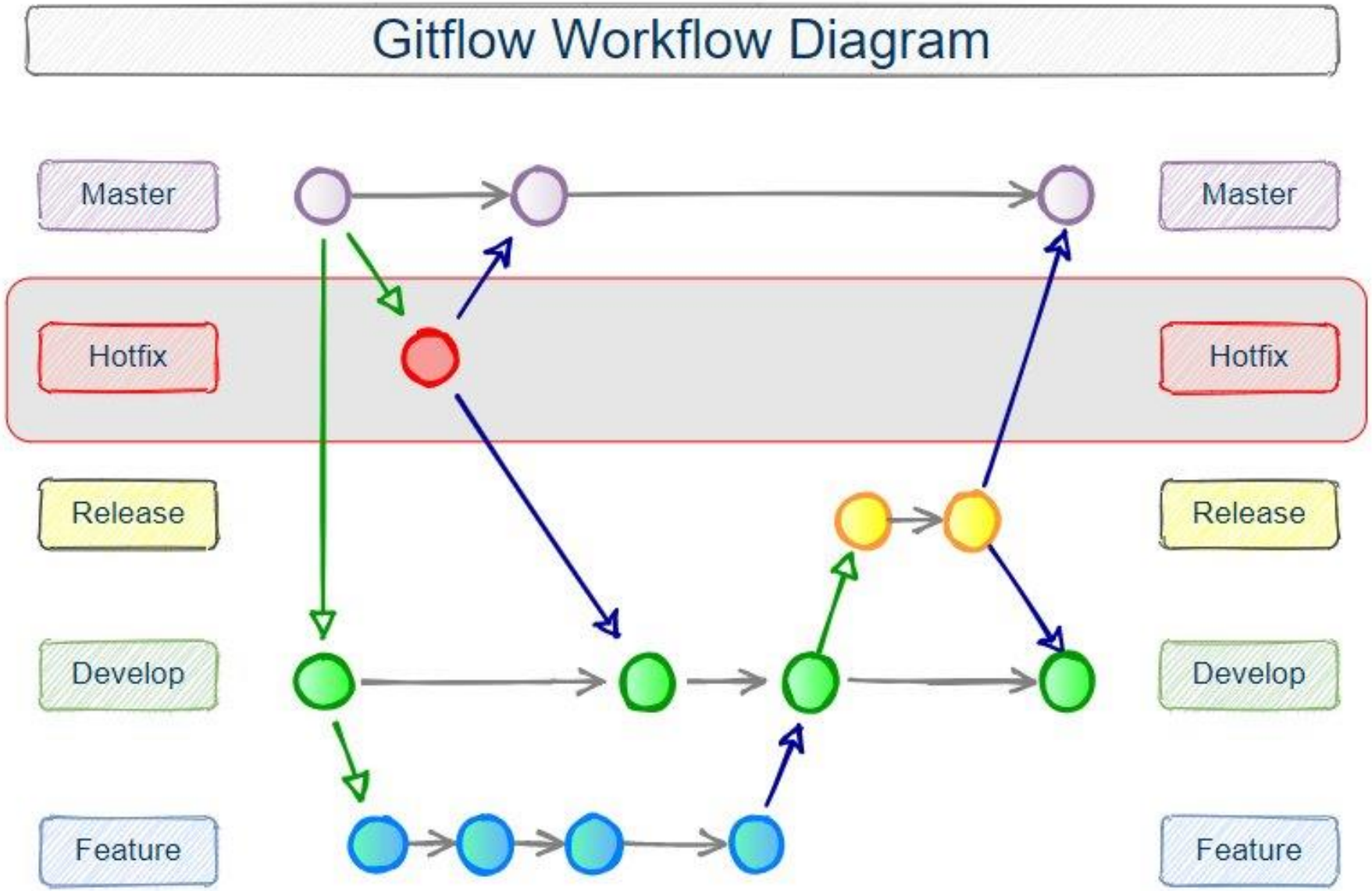

Server Computer
Version Database
Version 3
Version 2
Version 1
Computer A
File
Version Database
Version 3
Version 2
Version 1
Computer B
File
Version Database
Version 3
Version 2
Version 1



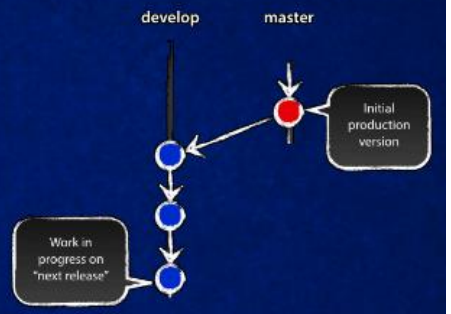

git


Local
Remote
Index
HEAD
HEAD
working directory
staging area
local repository
remote repo
git add
git commit
git push
git pull
git checkout
git merge


feature
BRANCH
feature
feature
BRANCH
feature
trunk
trunk
trunk
trunk
trunk
feature
BRANCH
feature


Gitflow Workflow Diagram
Master
Hotfix
Release
Develop
Feature
Master
Hotfix
Release
Develop
Feature

DEVELOP



Git

```
# Start develop branch
git checkout -b develop master
```

Git Flow

```
# Initialize flow
git flow init
```

MASTER



Git

```
# New repo
git init

# Clone repo
git clone git@server.name:repo.git
```

FEATURE/*

Git

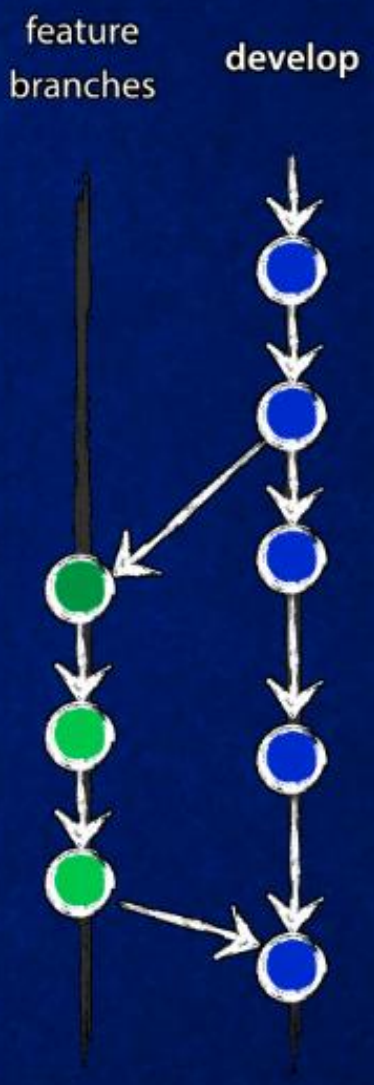
```
# Creación
git checkout -b feature/lorem-ipsum develop

# Finalización
git checkout develop
git merge --no-ff feature/lorem-ipsum
git branch -d feature/lorem-ipsum
git push origin develop
```

Git Flow

```
# Creación
git flow feature start lorem-ipsum

# Finalización
git flow feature finish lorem-ipsum
```



RELEASE/‹VERSION›

Git

```
# Creación
git checkout -b release/v2.3.0 develop

# Finalización
git checkout master
git merge --no-ff release/v2.3.0
git tag -a v2.3.0

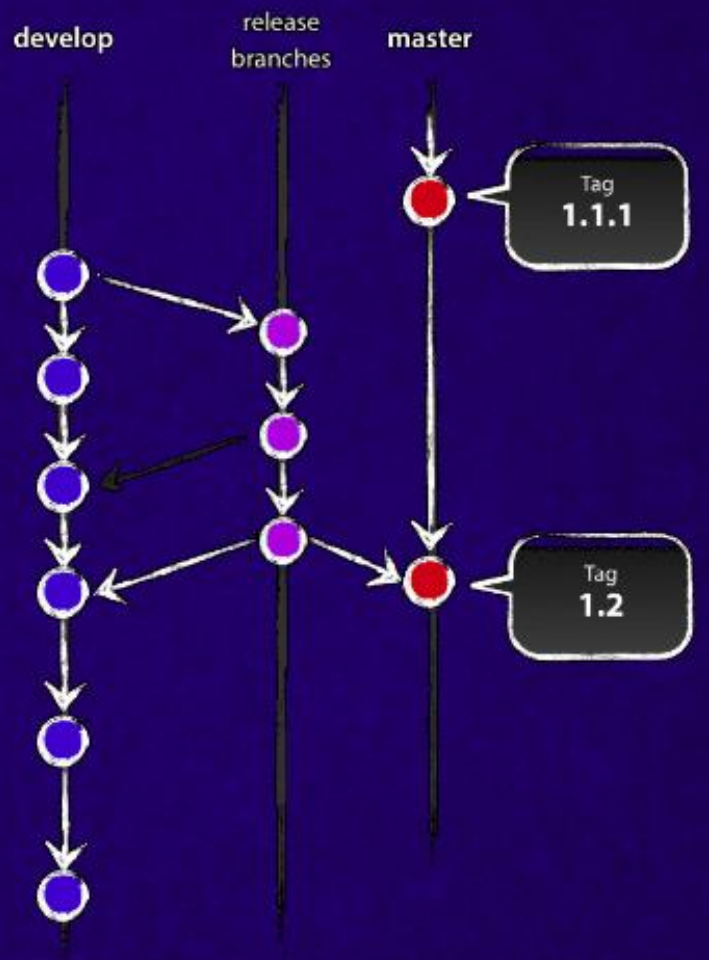
git checkout develop
git merge --no-ff release/v2.3.0

git branch -d release/v2.3.0
```

Git Flow

```
# Creación
git flow release start v2.3.0

# Finalización
git flow release finish v2.3.0
```



HOTFIX/‹VERSION›

Git

```
# Creación
git checkout -b hotfix/v2.3.7 master

# Finalización
git checkout master
git merge --no-ff hotfix/v2.3.7
git tag -a v2.3.7

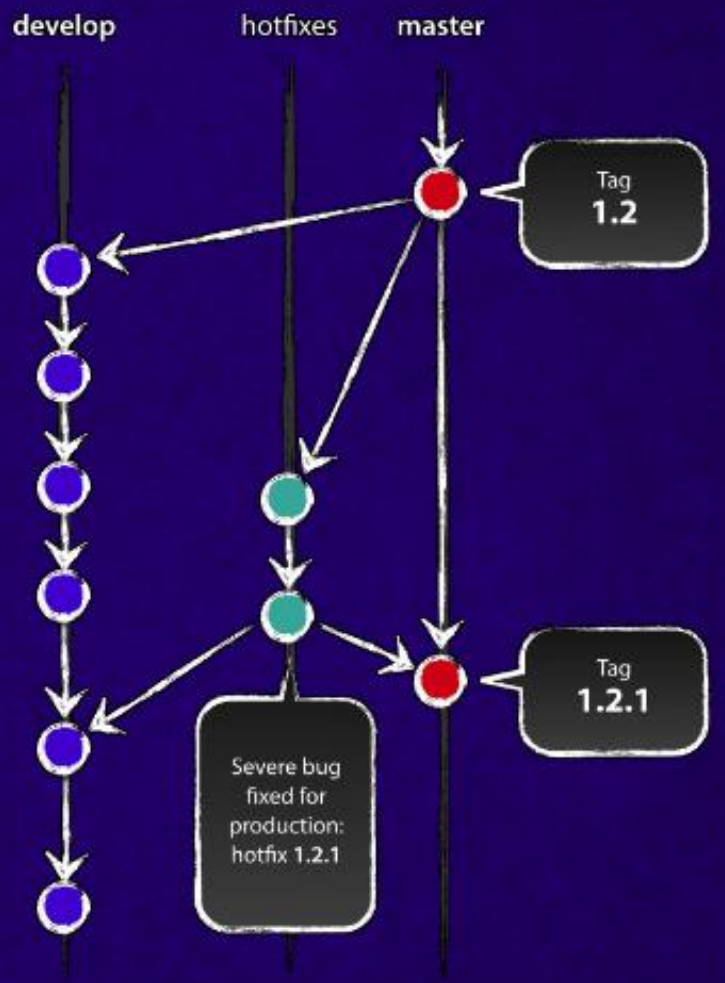
git checkout develop
git merge --no-ff hotfix/v2.3.7

git branch -d hotfix/v2.3.7
```

Git Flow

```
# Creación
git flow hotfix start v2.3.7

# Finalización
git flow hotfix finish v2.3.7
```



EXERCISE 1

Advanced Git Concepts

Rebasing: Rewrites history to create a linear commit timeline.

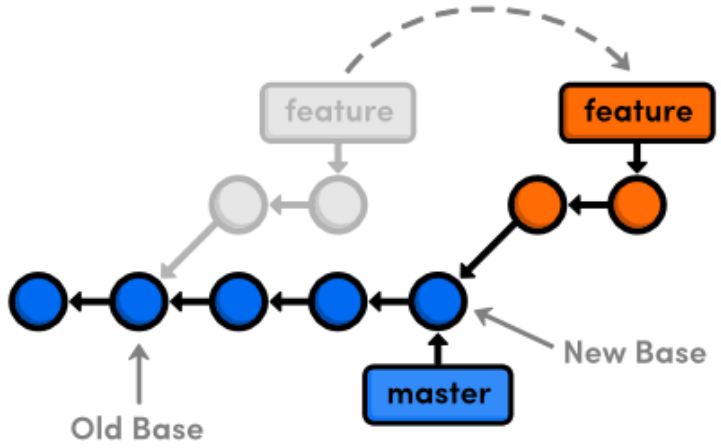
Cherry-pick: Applies specific commits from one branch to another.

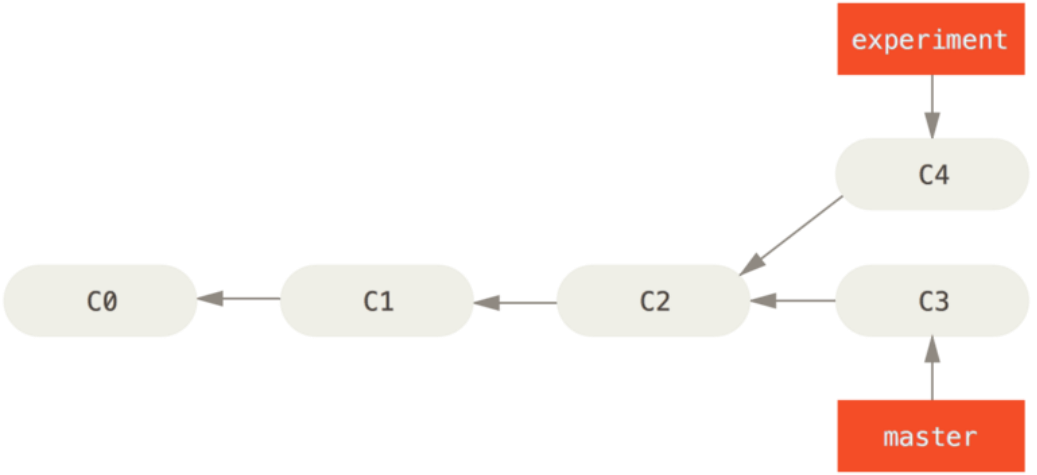
Stashing: Temporarily saves changes without committing.

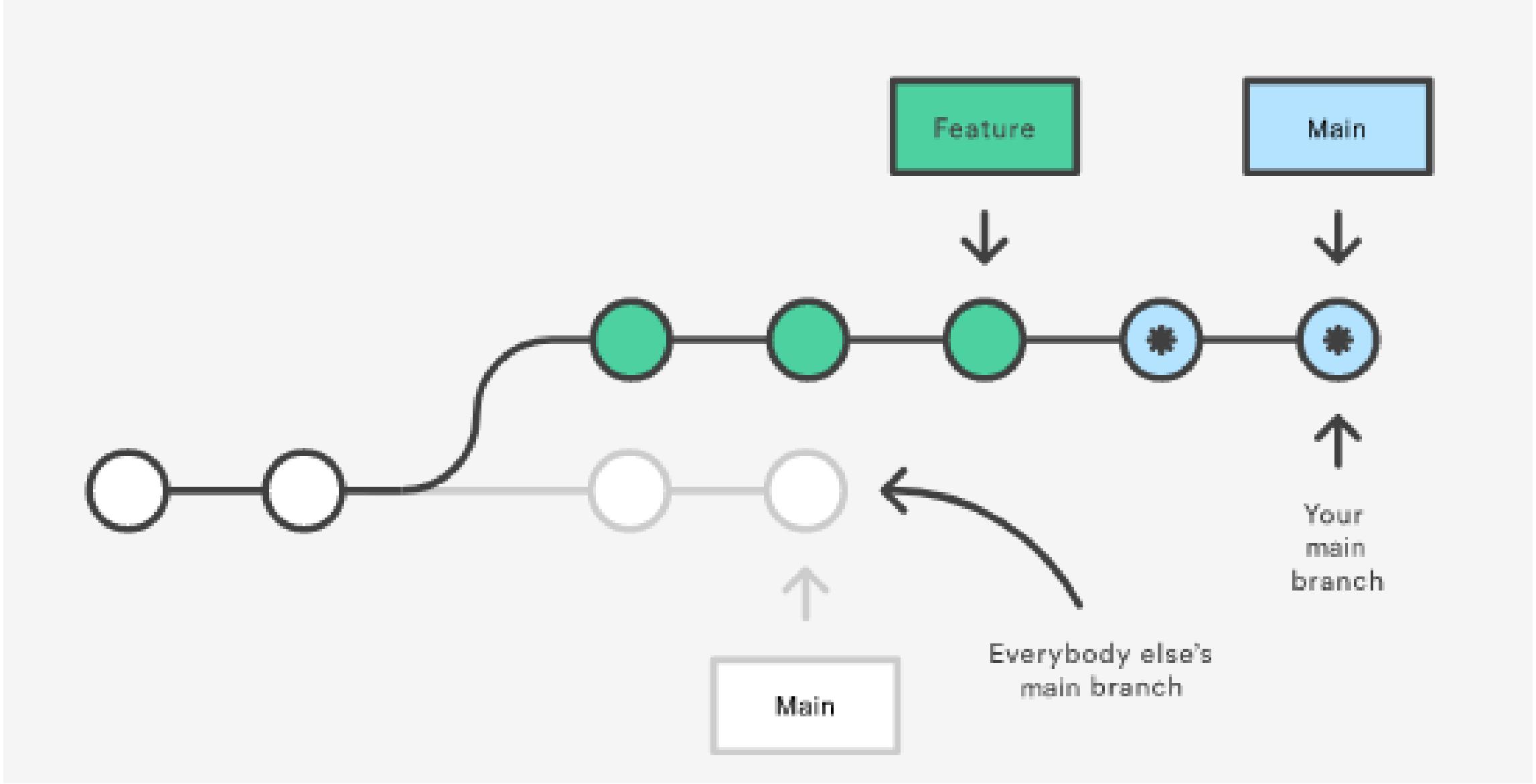
Hooks: Custom scripts triggered by Git events (e.g., pre-commit).

Submodules: Includes one repository as a subdirectory of another.

REBASE


feature
feature
New Base
master
Old Base


experiment
C4
C0
C1
C2
C3
master

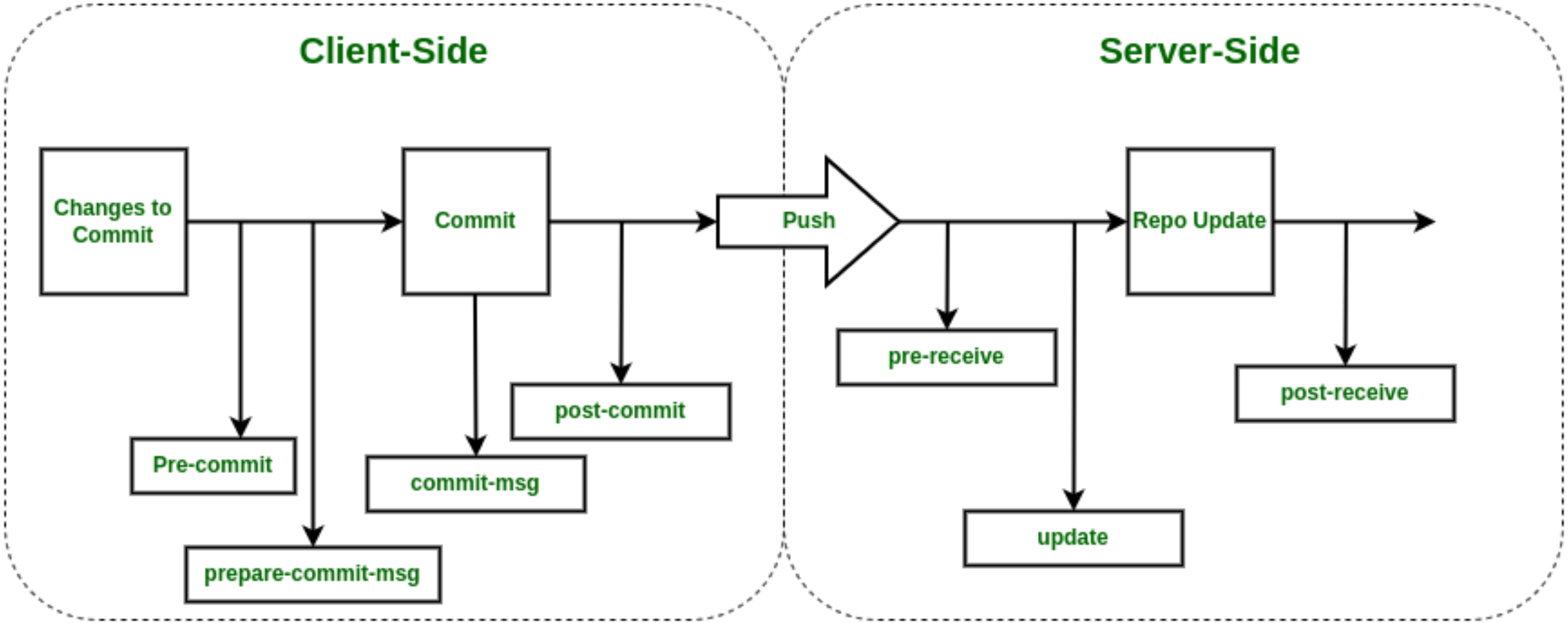

Feature
Main
Your
main
branch
Main
Everybody else's
main branch

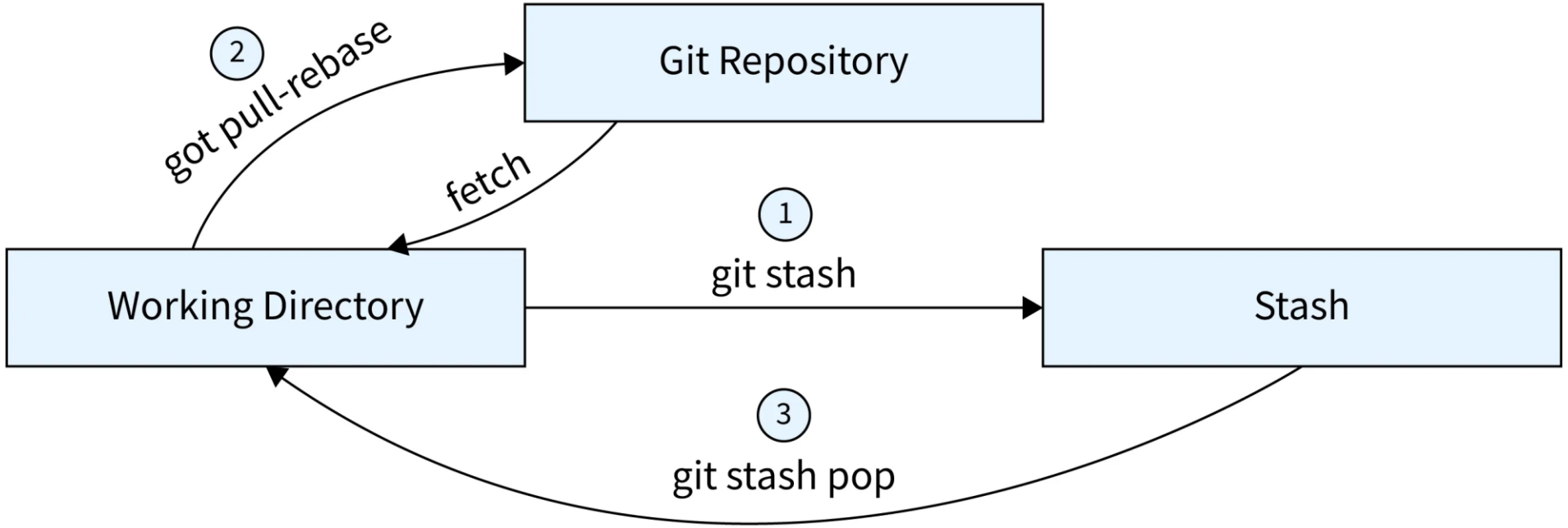
DEMO



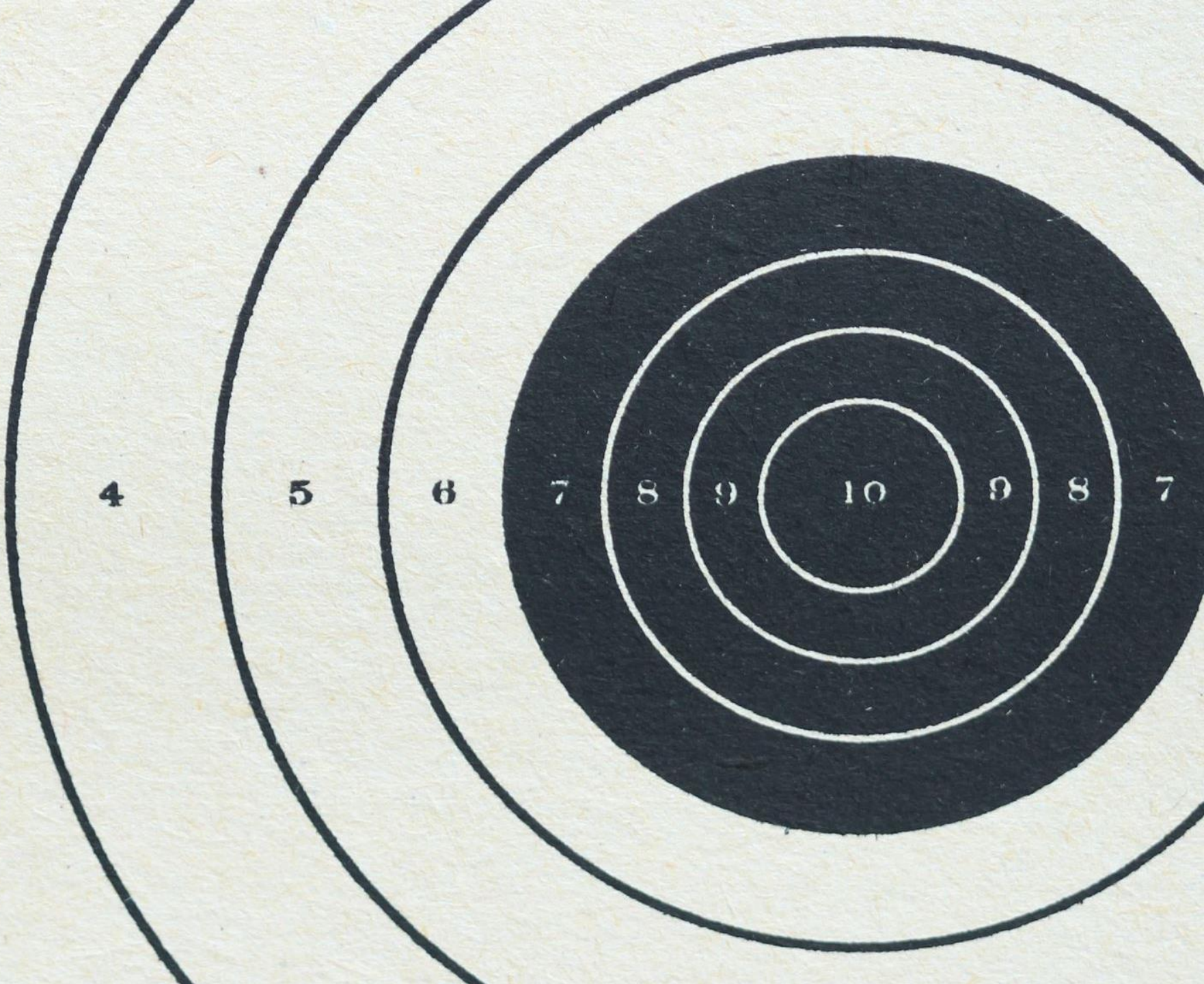
HOOKS


HOSTINGER
Tres. Dos. Online
Acción del usuario
1
Cambios en Stage
2
Cambios de Commit
3
Ingreso del Mensaje de Commit
4
¡Todo hecho!
Hooks
Notificación Pre-Commit
Notificación Post-Commit

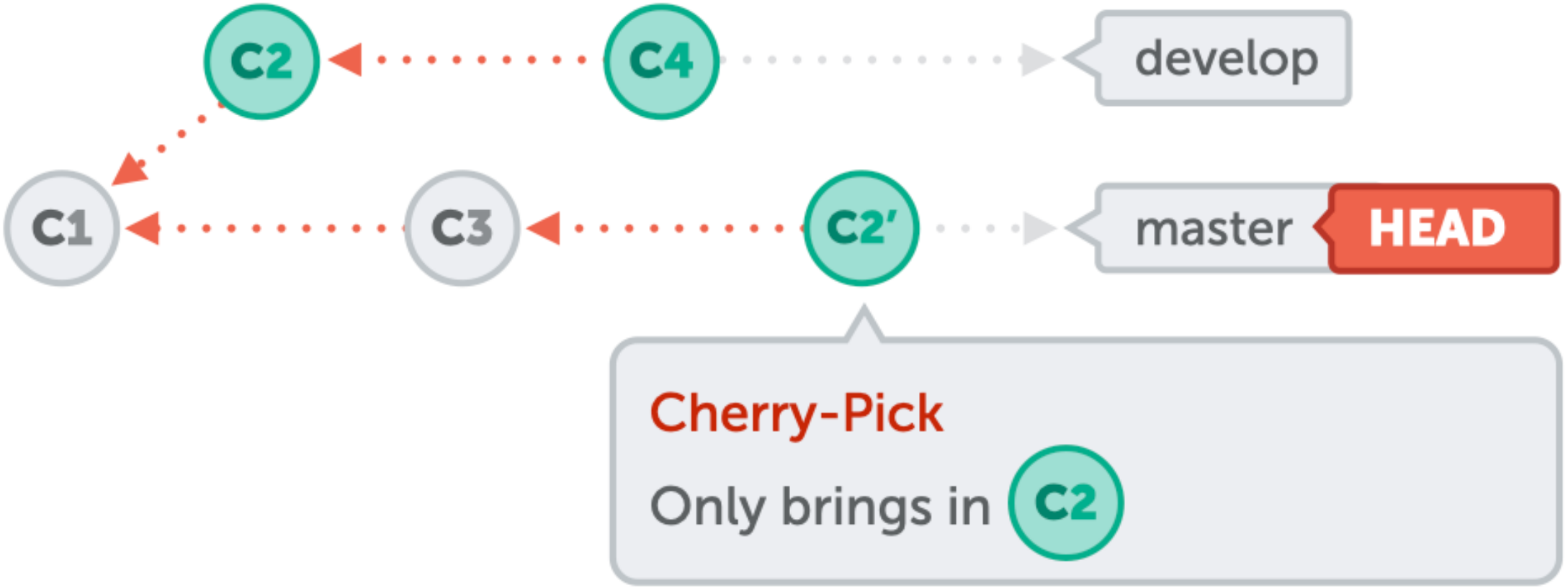

Client-Side
Server-Side
Changes to Commit
Commit
Push
Repo Update
Pre-commit
commit-msg
post-commit
prepare-commit-msg
pre-receive
update
post-receive

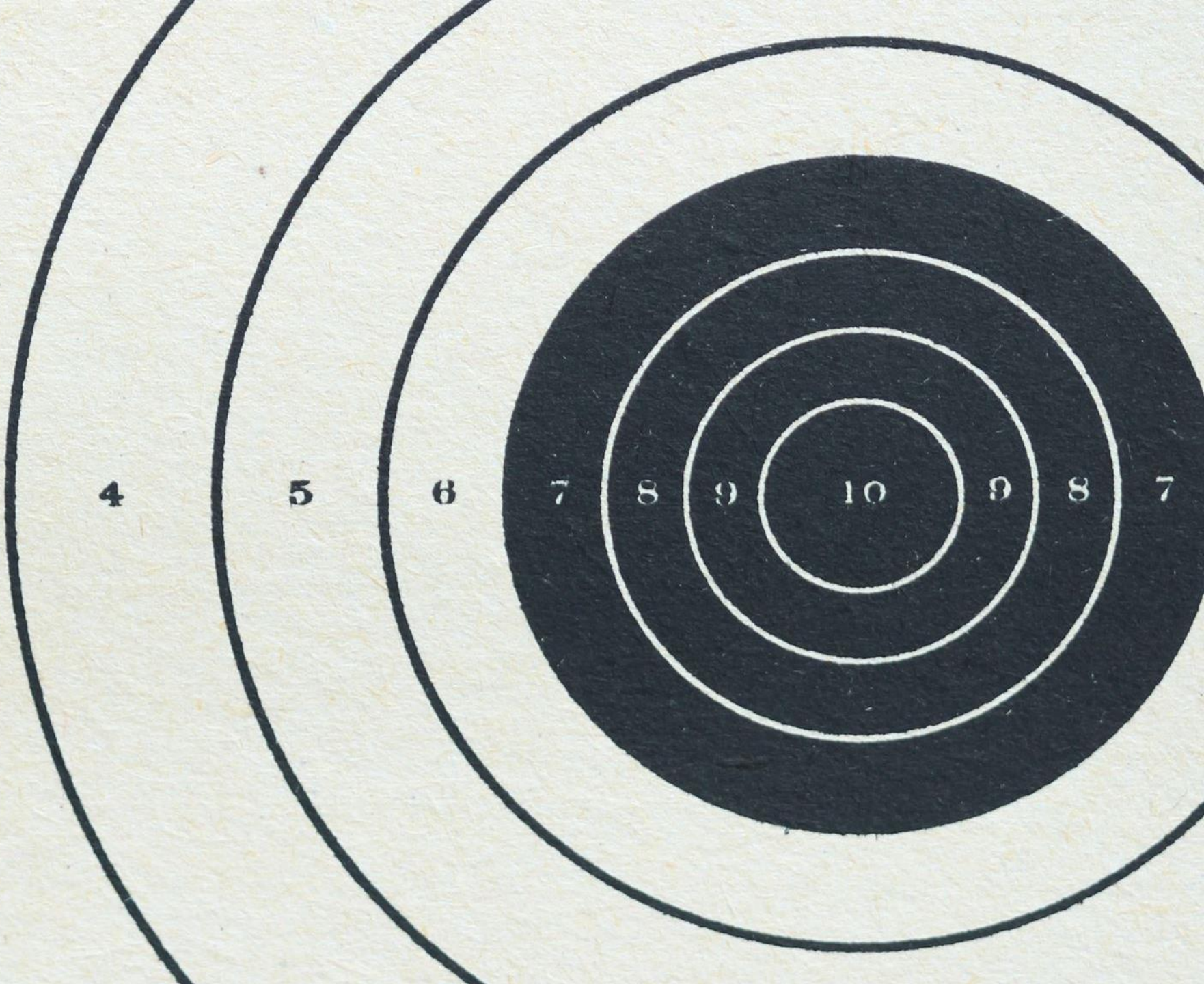


SCALER Topics



DEMO

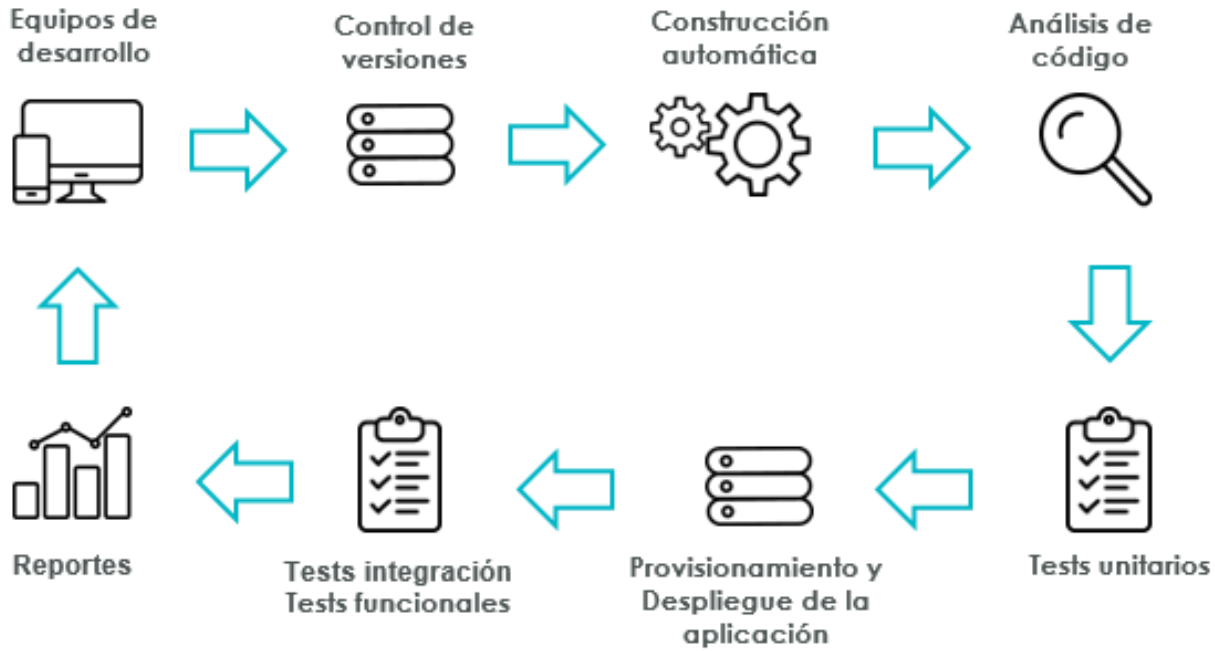

C2
C4
develop
C1
C3
C2'
master
HEAD
Cherry-Pick
Only brings in
C2

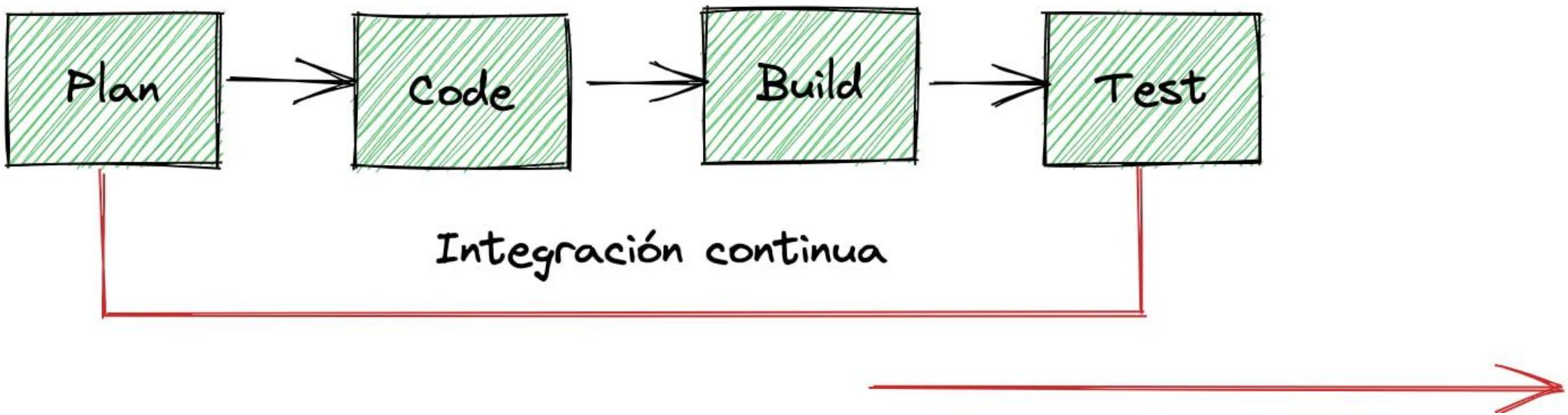


DEMO

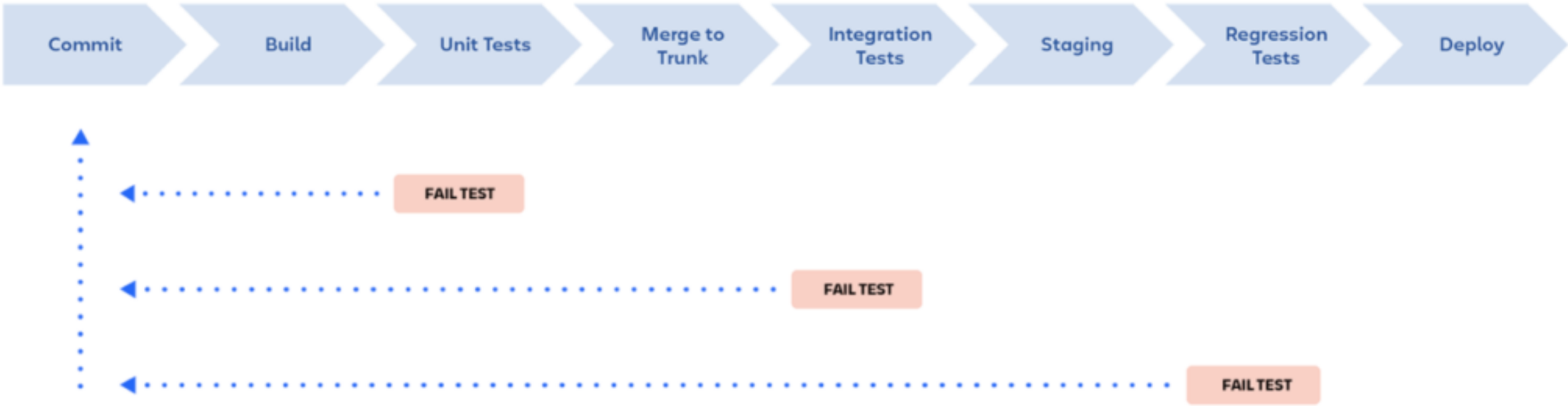
EXERCISE 2

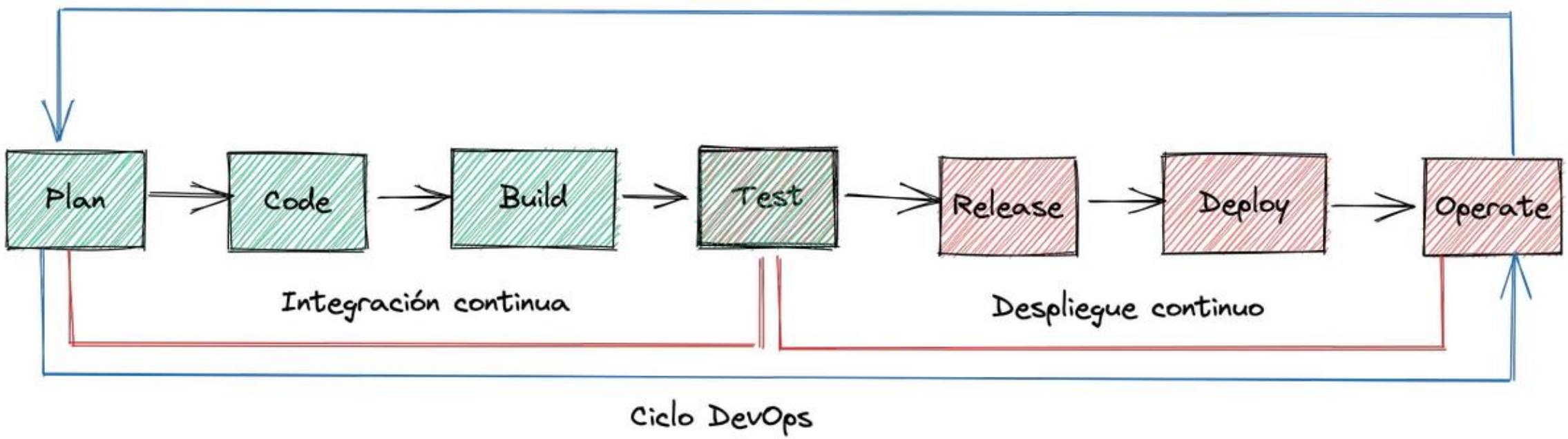

DevOps
INTEGRACIÓN CONTINUA
6 BENEFICIOS
1
Mejora la calidad del código
2
Detección de errores más rápida y fácil
3
Reduce tareas repetitivas y manuales
4
Puede crear versiones de pruebas en cualquier momento
5
Completa visibilidad del proyecto
6
Mayor confianza y seguridad del equipo de trabajo


Equipos de desarrollo
Control de versiones
Construcción automática
Análisis de código
Tests unitarios
Provisionamiento y Despliegue de la aplicación
Tests integración
Tests funcionales
Reportes

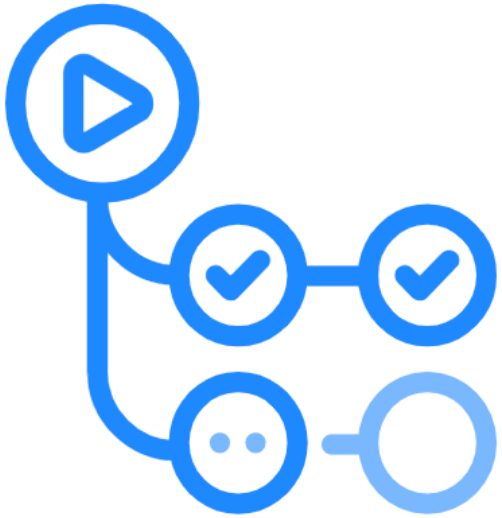

Plan
Code
Build
Test
Integración continua

PIPELINE


Commit
Build
Unit Tests
Merge to Trunk
Integration Tests
Staging
Regression Tests
Deploy
FAIL TEST
FAIL TEST
FAIL TEST

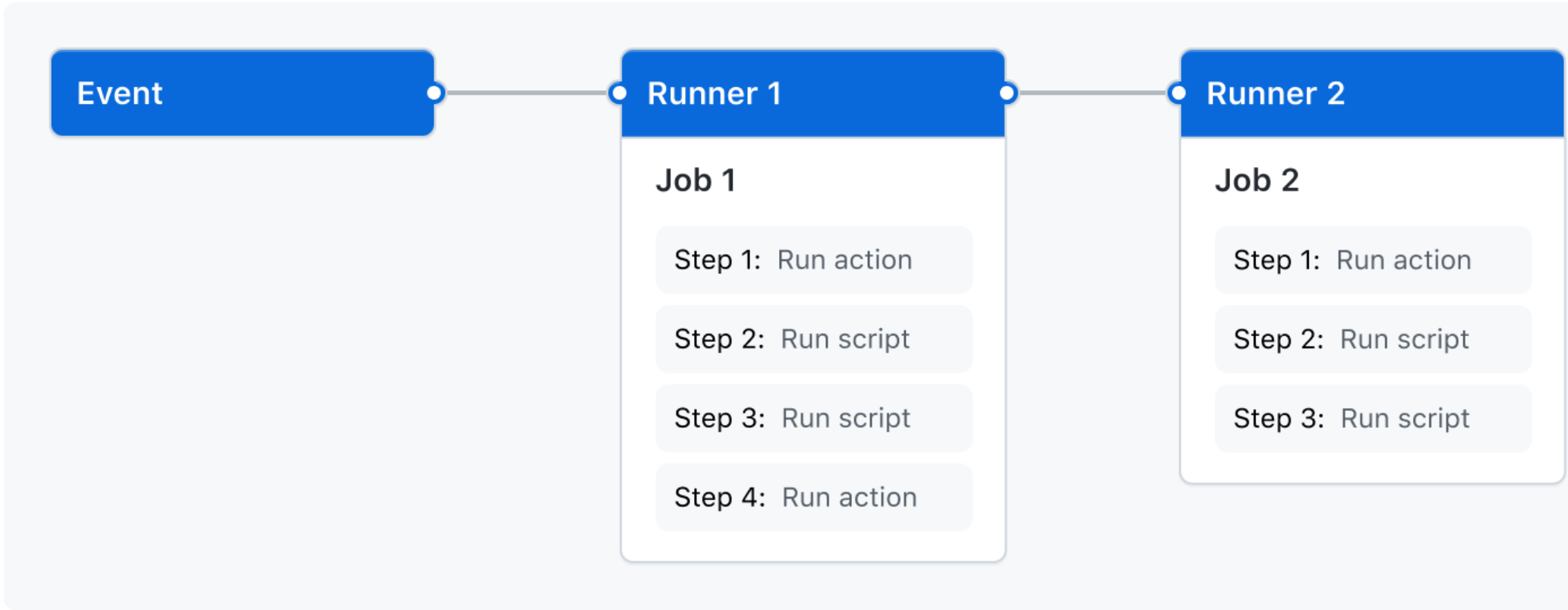


Ciclo DevOps



GitHub Actions

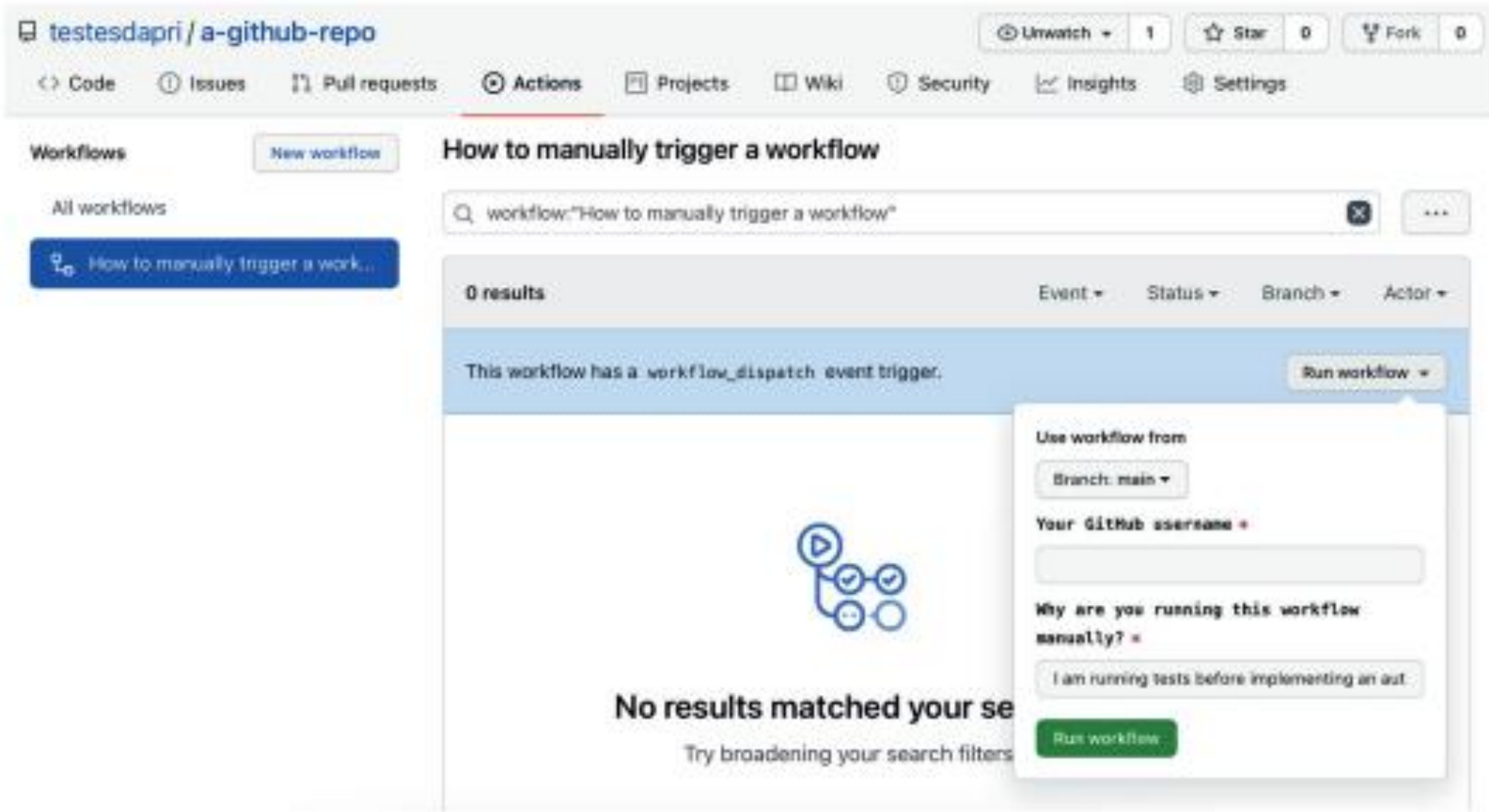
Github actions


Event
Runner 1
Job 1
Step 1: Run action
Step 2: Run script
Step 3: Run script
Step 4: Run action
Runner 2
Job 2
Step 1: Run action
Step 2: Run script
Step 3: Run script

```yaml
name: Nombre-del-Workflow

on:
  evento:
    - evento-trigger

jobs:
  nombre-del-trabajo:
    runs-on: tipo-de-maquina
    steps:
      - nombre-del-paso:
          run: comando-a-ejecutar
```


testesdapri / a-github-repo
Unwatch 1
Star 0
Fork 0
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Workflows
New workflow
All workflows
How to manually trigger a work...
How to manually trigger a workflow
workflow:"How to manually trigger a workflow"
0 results
Event
Status
Branch
Actor
This workflow has a workflow_dispatch event trigger.
Run workflow
Use workflow from
Branch: main
Your GitHub username *
Why are you running this workflow manually? *
I am running tests before implementing an aut
Run workflow
No results matched your se
Try broadening your search filters

YAML

```yaml
name: Greeting on variable day

on:
  workflow_dispatch

env:
  DAY_OF_WEEK: Monday

jobs:
  greeting_job:
    runs-on: ubuntu-latest
    env:
      Greeting: Hello
    steps:
      - name: "Say Hello Mona it's Monday"
        run: echo "$Greeting $First_Name. Today is $DAY_OF_WEEK!"
        env:
          First_Name: Mona
```

Actions secrets and variables

New repository secret

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. Learn more about encrypted secrets. Variables are shown as plain text and are used for **non-sensitive** data. Learn more about variables.

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets | Variables

Actions

An entirely new way to automate your development workflow.

18381 results filtered by Actions ×

Actions



Setup Java JDK

By actions

Set up a specific version of the Java JDK and add the command-line tools to the PATH

☆ 1.1k stars



Setup .NET Core SDK

By actions

Used to build and publish .NET source. Set up a specific version of the .NET and authentication to private NuGet repository

☆ 747 stars



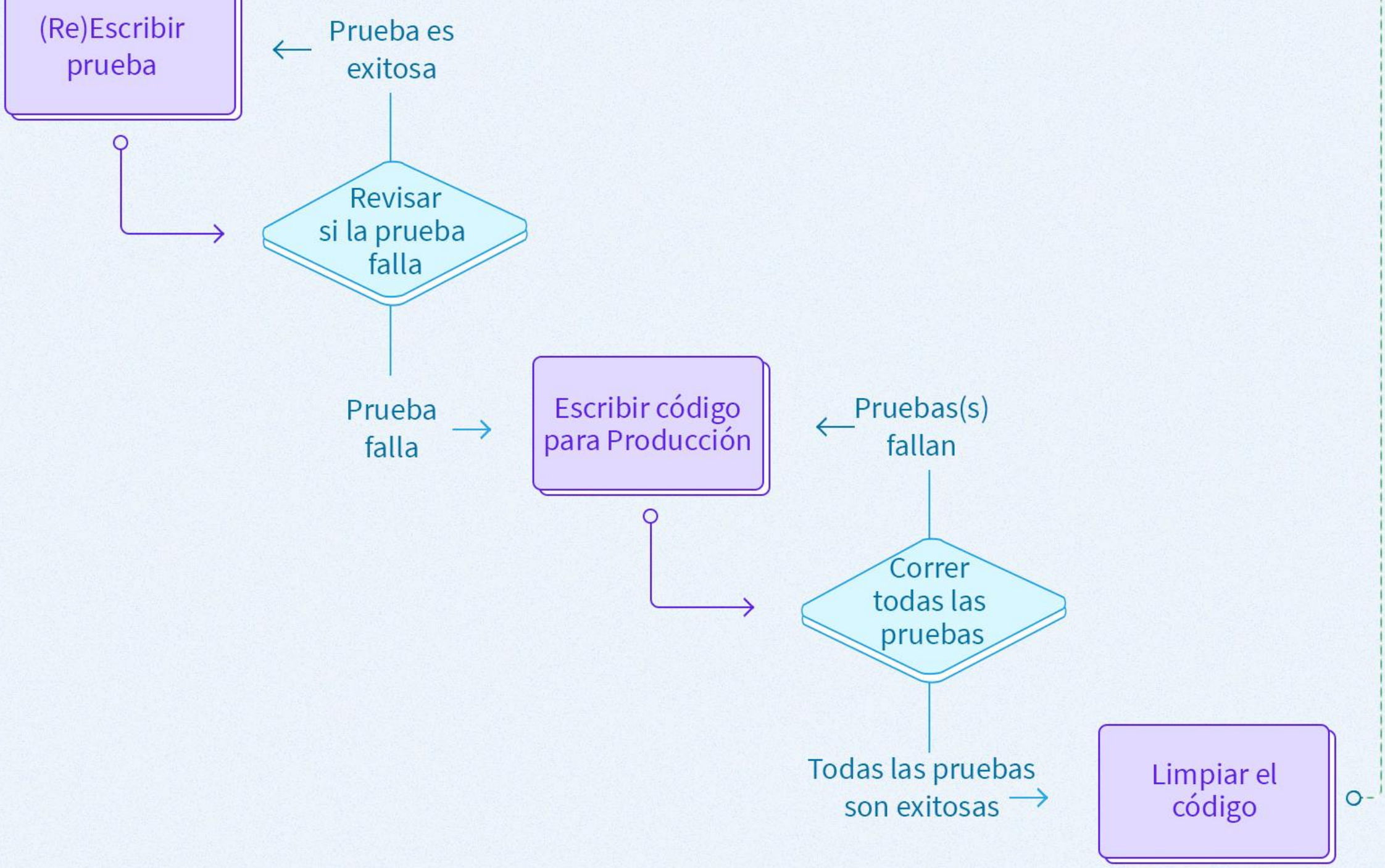


vs.

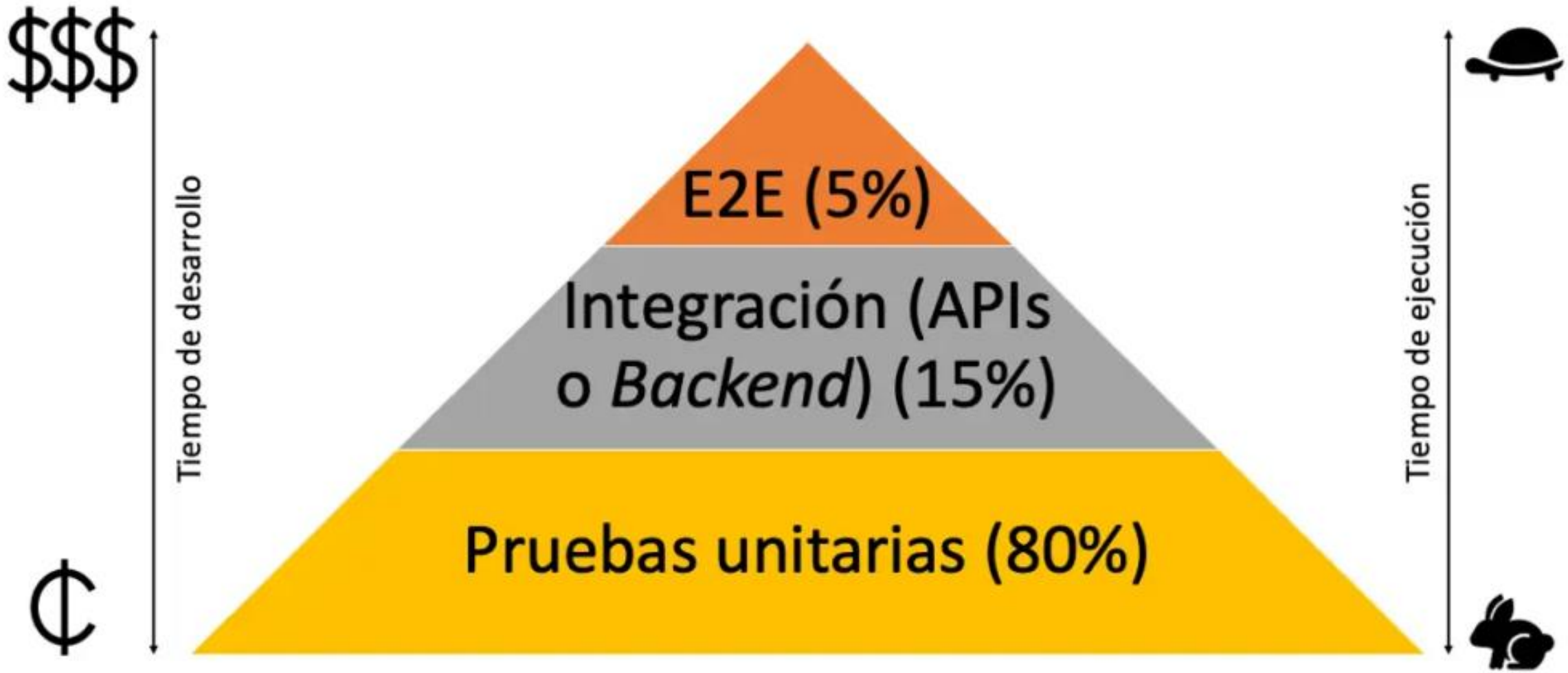


Pruebas dinámicas

Pruebas estáticas

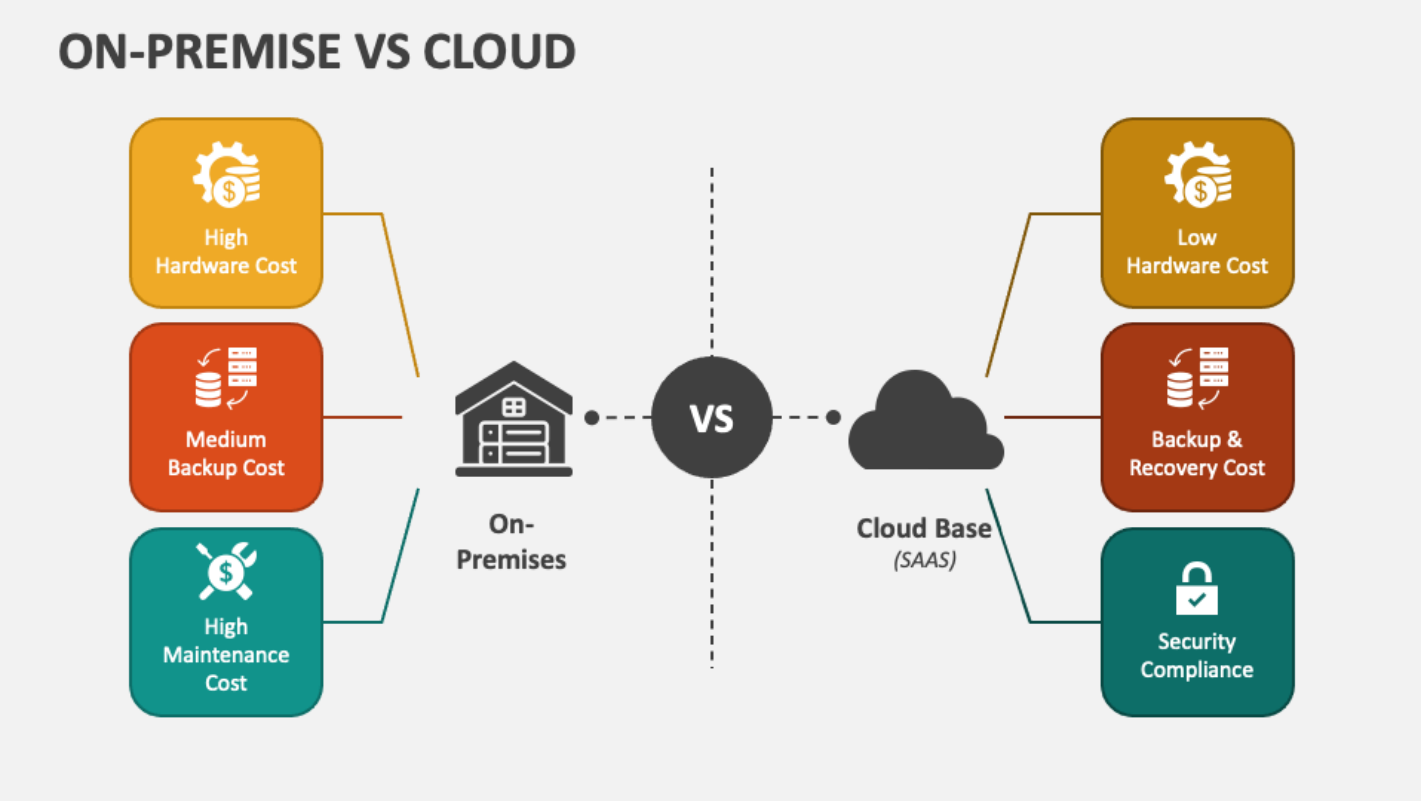

(Re)Escribir prueba
Prueba es exitosa
Revisar si la prueba falla
Prueba falla
Escribir código para Producción
Pruebas(s) fallan
Correr todas las pruebas
Todas las pruebas son exitosas
Limpiar el código




$$$
¢
Tiempo de desarrollo
E2E (5%)
Integración (APIs o *Backend*) (15%)
Pruebas unitarias (80%)
Tiempo de ejecución

EXERCISE 3

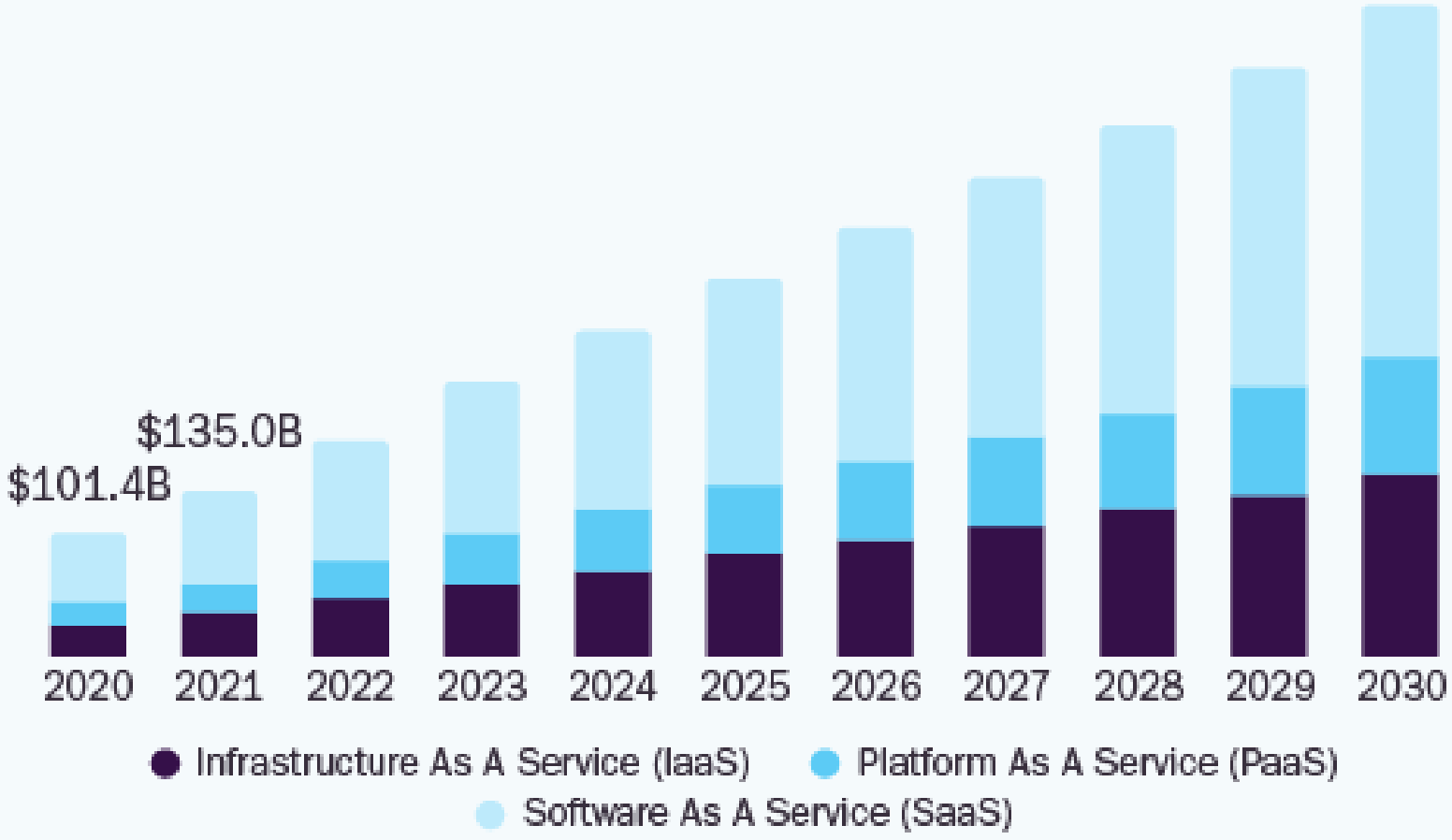
CLOUD COMPUTING VS ON-PREMISE




Data Security
Scalability
Disaster Recovery
Mobility
Reduced Costs
Cloud Computing

U.S. Cloud Computing Market

size, by service, 2020 - 2030 (USD Billion)

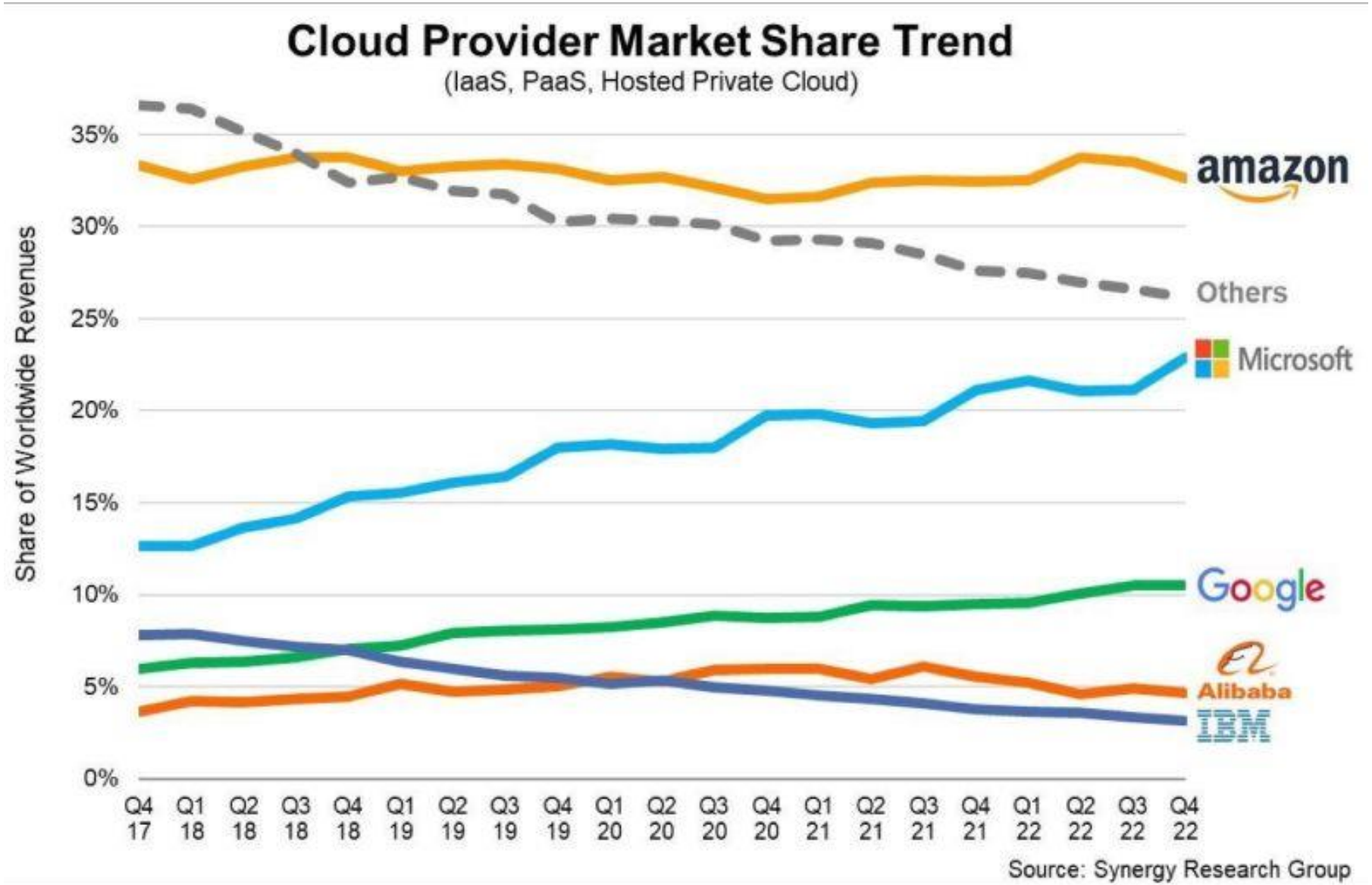


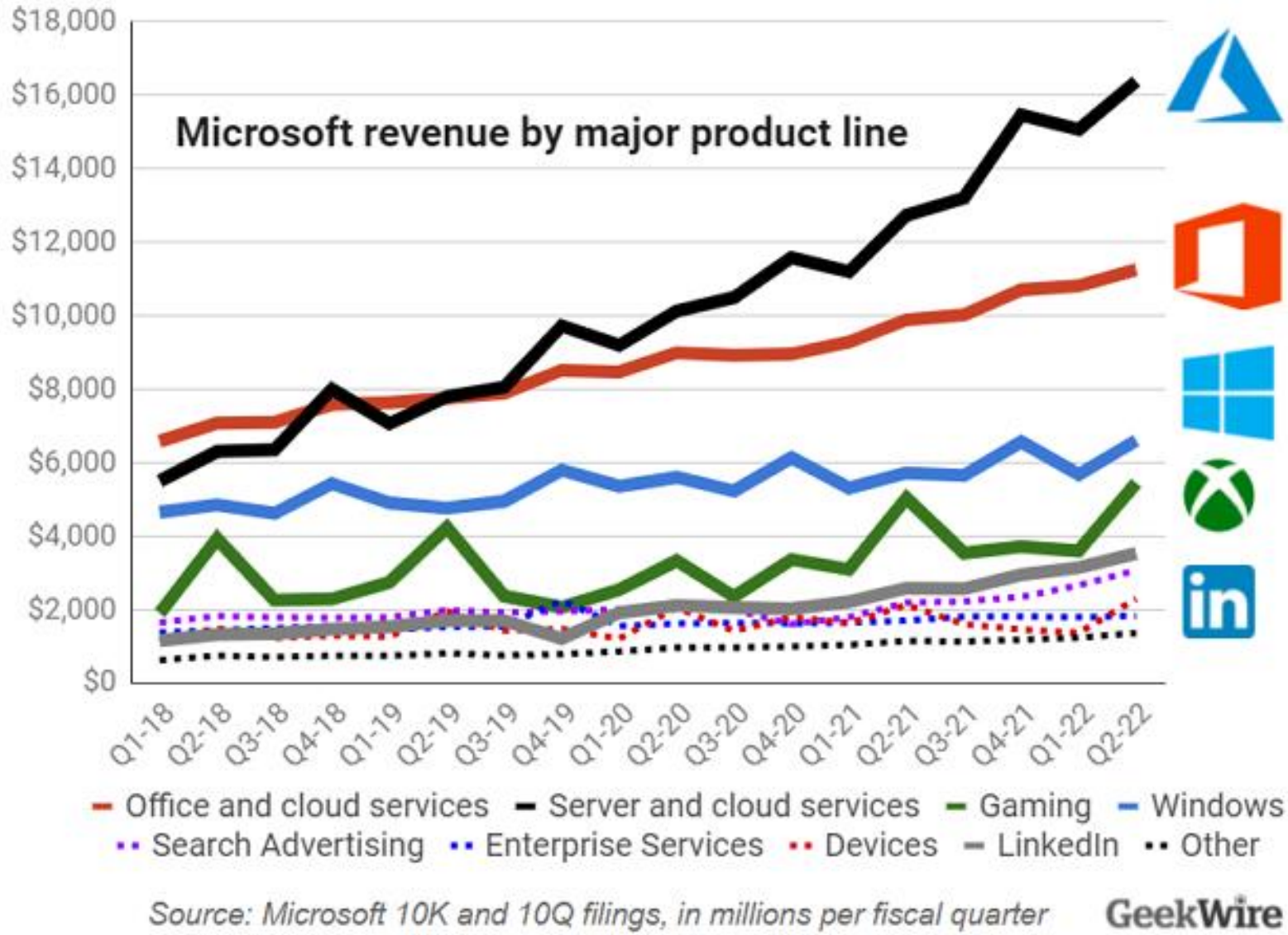
GRAND VIEW RESEARCH

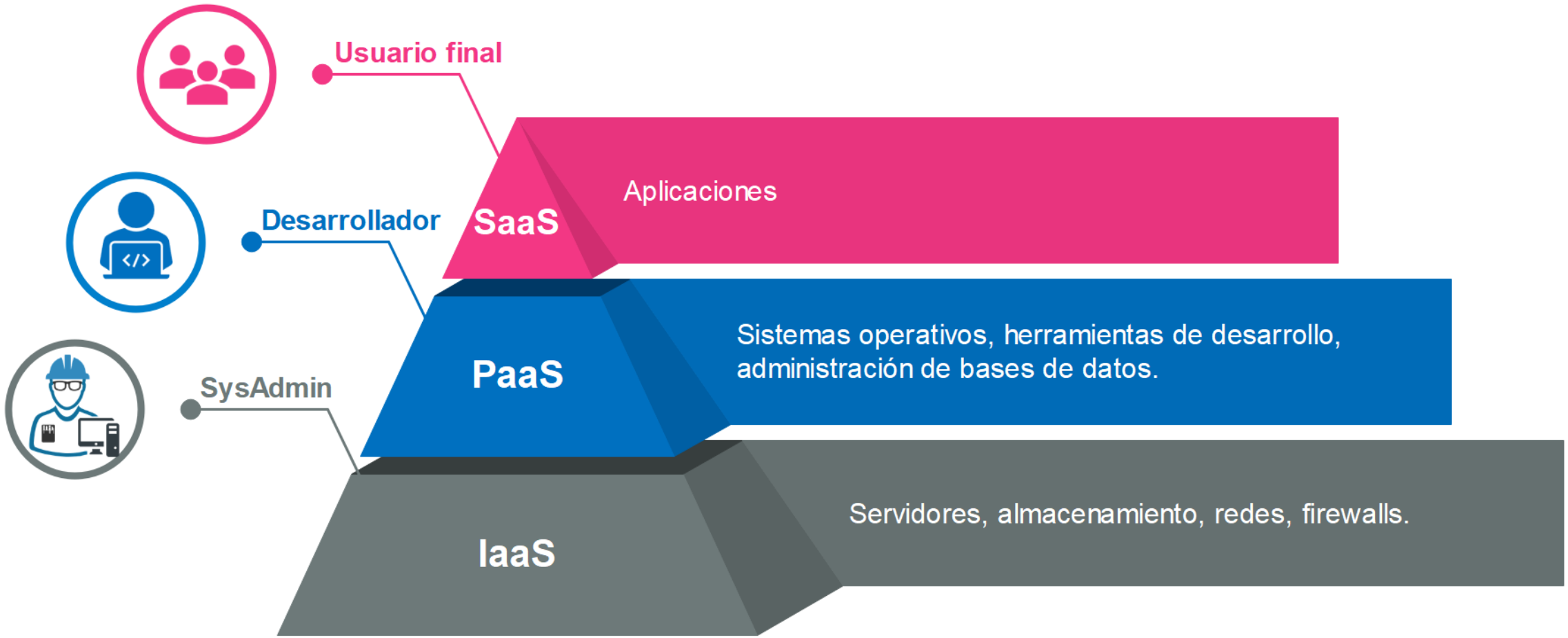
13.1%

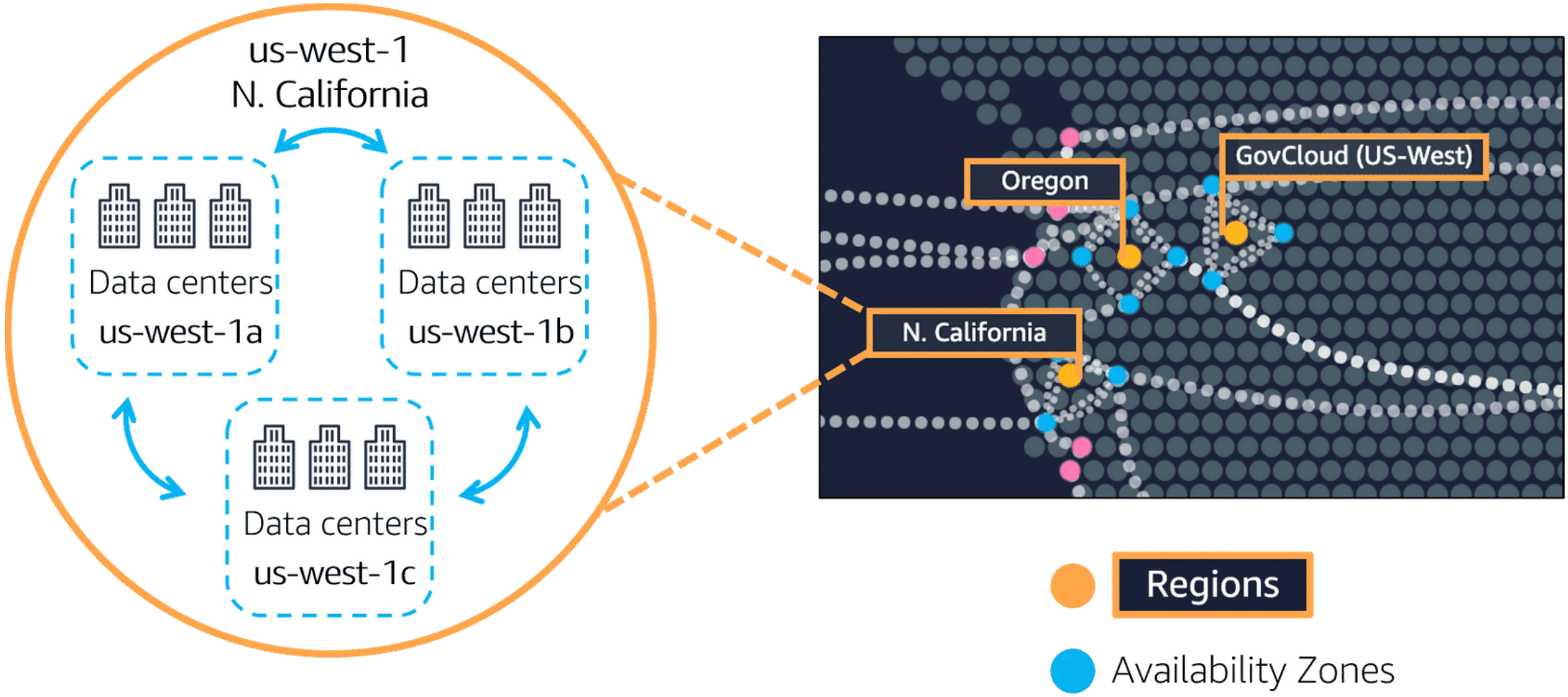
U.S. Market CAGR, 2023 - 2030

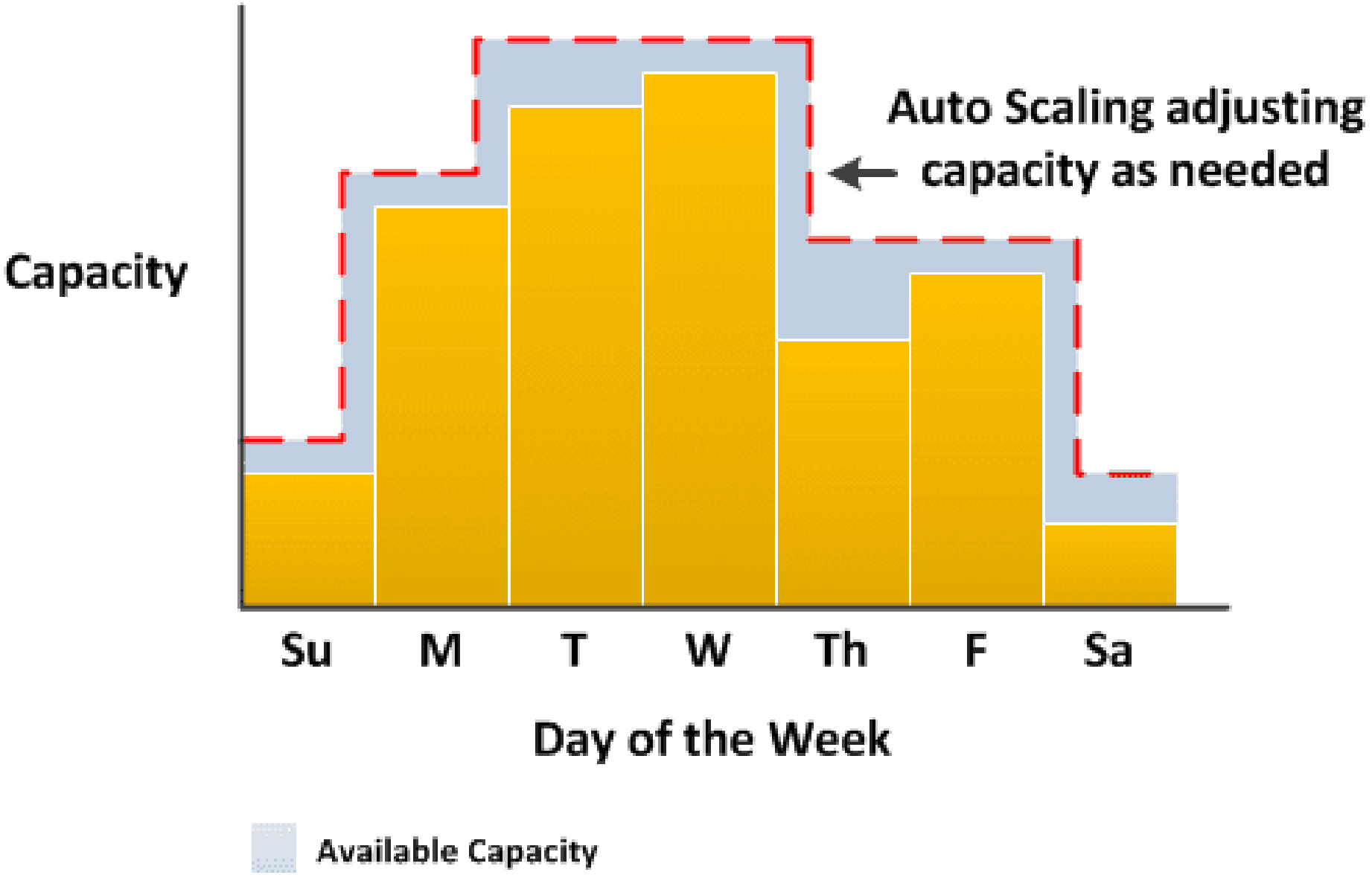
Source:
www.grandviewresearch.com


Cloud Provider Market Share Trend
(IaaS, PaaS, Hosted Private Cloud)
Share of Worldwide Revenues
35%
30%
25%
20%
15%
10%
5%
0%
Q4 17
Q1 18
Q2 18
Q3 18
Q4 18
Q1 19
Q2 19
Q3 19
Q4 19
Q1 20
Q2 20
Q3 20
Q4 20
Q1 21
Q2 21
Q3 21
Q4 21
Q1 22
Q2 22
Q3 22
Q4 22
amazon
Others
Microsoft
Google
Alibaba
IBM
Source: Synergy Research Group


Microsoft revenue by major product line
$18,000
$16,000
$14,000
$12,000
$10,000
$8,000
$6,000
$4,000
$2,000
$0
Q1-18
Q2-18
Q3-18
Q4-18
Q1-19
Q2-19
Q3-19
Q4-19
Q1-20
Q2-20
Q3-20
Q4-20
Q1-21
Q2-21
Q3-21
Q4-21
Q1-22
Q2-22
Office and cloud services
Server and cloud services
Gaming
Windows
Search Advertising
Enterprise Services
Devices
LinkedIn
Other
Source: Microsoft 10K and 10Q filings, in millions per fiscal quarter
GeekWire

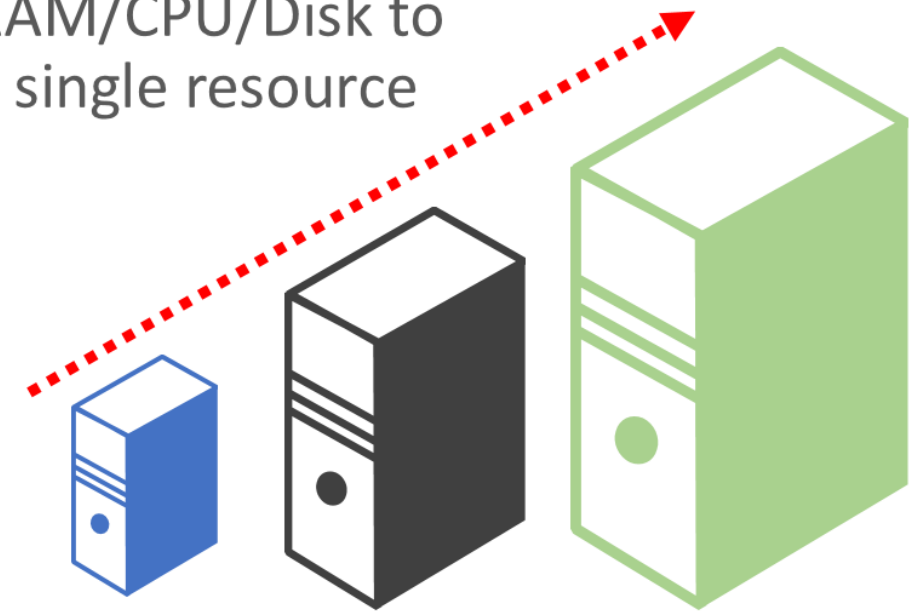

Usuario final
Desarrollador
SysAdmin
SaaS
Aplicaciones
PaaS
Sistemas operativos, herramientas de desarrollo, administración de bases de datos.
IaaS
Servidores, almacenamiento, redes, firewalls.


us-west-1
N. California
Data centers
us-west-1a
Data centers
us-west-1b
Data centers
us-west-1c
Oregon
GovCloud (US-West)
N. California
Regions
Availability Zones


Capacity
Auto Scaling adjusting
capacity as needed
Su
M
T
W
Th
F
Sa
Day of the Week
Available Capacity

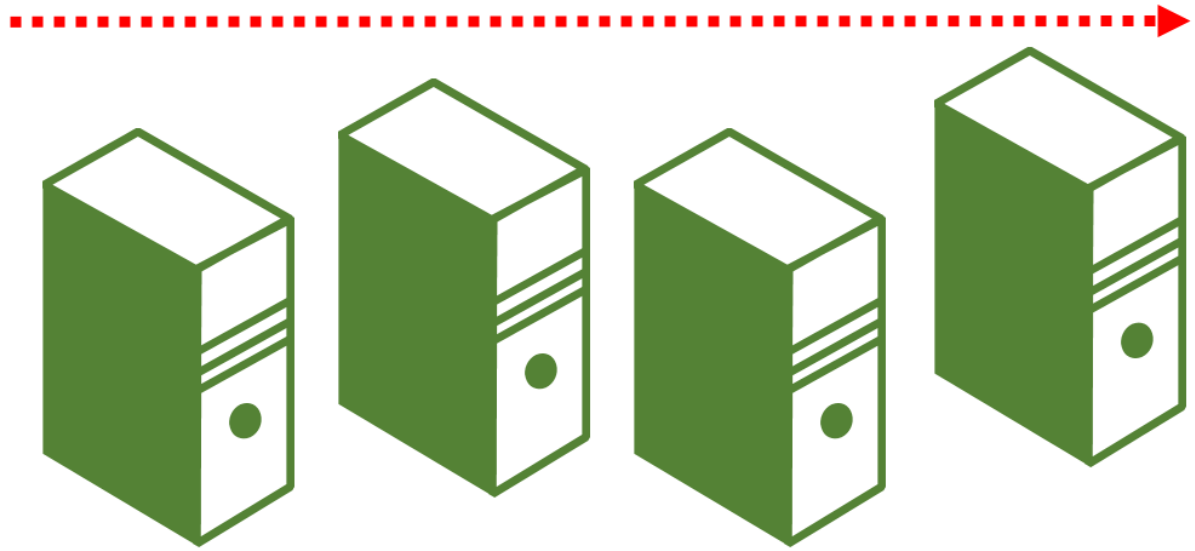
Scale Up (vertical scaling)

Increase capacity by adding RAM/CPU/Disk to a single resource



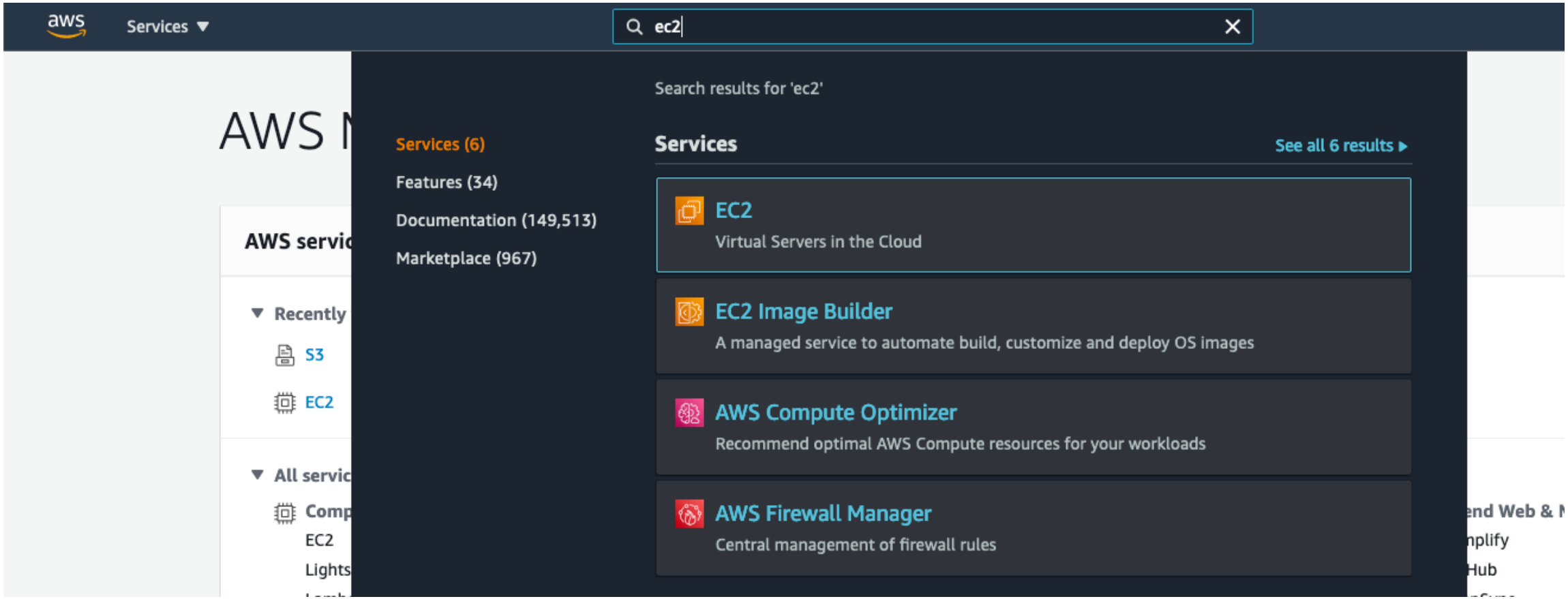
Scale Out (horizontal scaling)

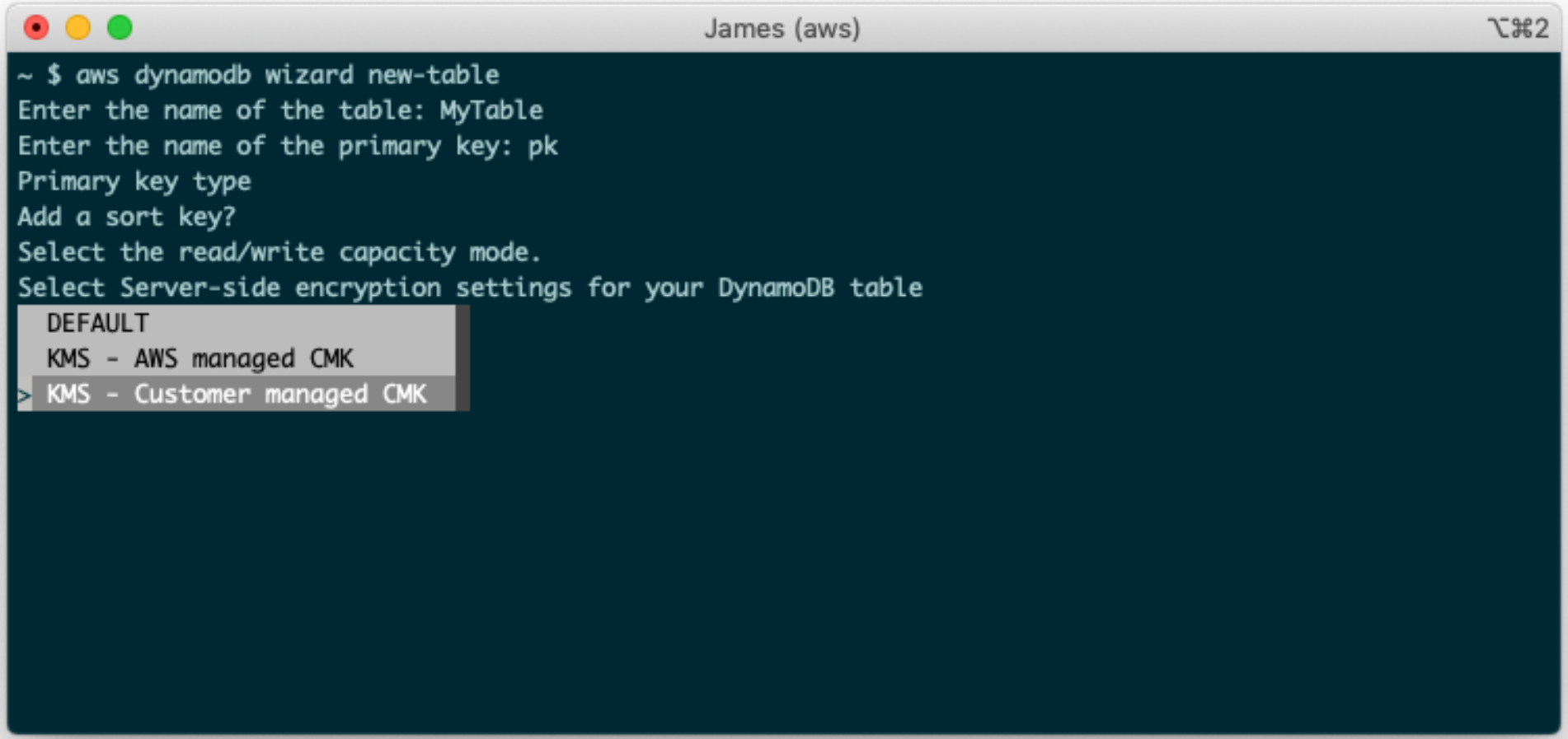
Increase capacity by adding resources

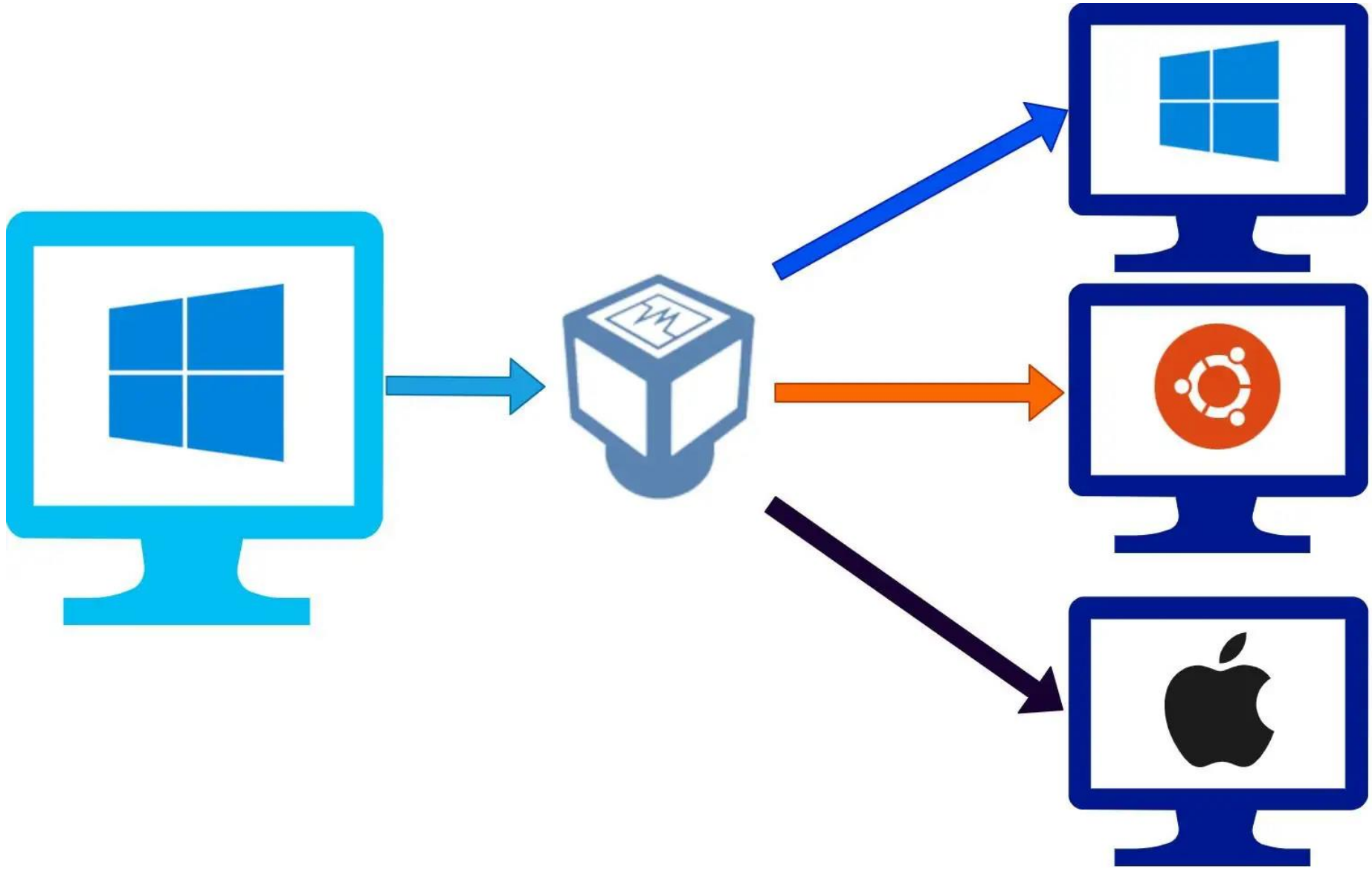



Microsoft
14:08
chat.openai.com
ChatGPT


aws

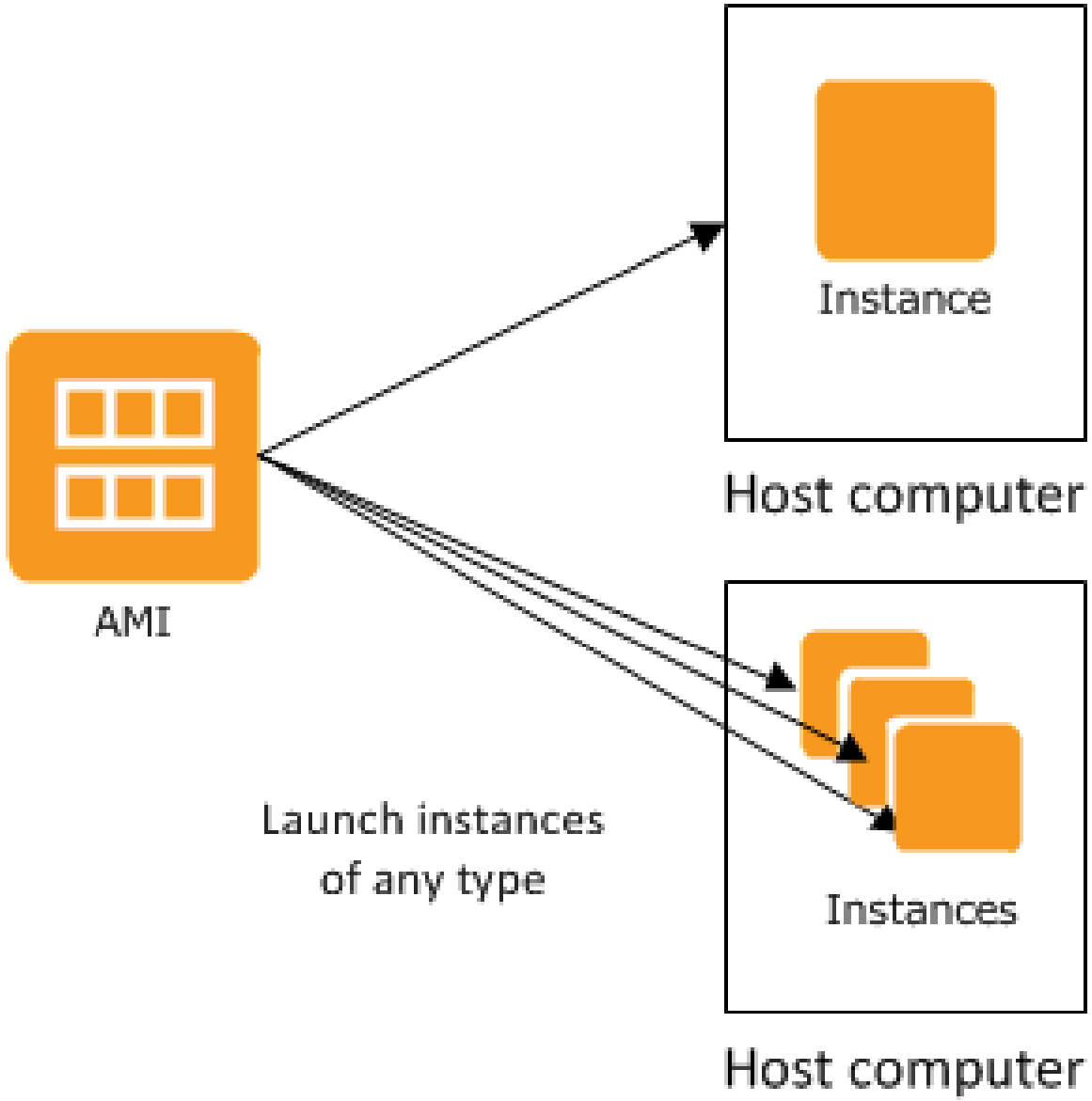

aws
Services
ec2
AWS M
AWS servic
Recently
S3
EC2
All servic
Comp
EC2
Lights
Search results for 'ec2'
Services (6)
Features (34)
Documentation (149,513)
Marketplace (967)
Services
See all 6 results
EC2
Virtual Servers in the Cloud
EC2 Image Builder
A managed service to automate build, customize and deploy OS images
AWS Compute Optimizer
Recommend optimal AWS Compute resources for your workloads
AWS Firewall Manager
Central management of firewall rules
nd Web & M
nplify
Hub

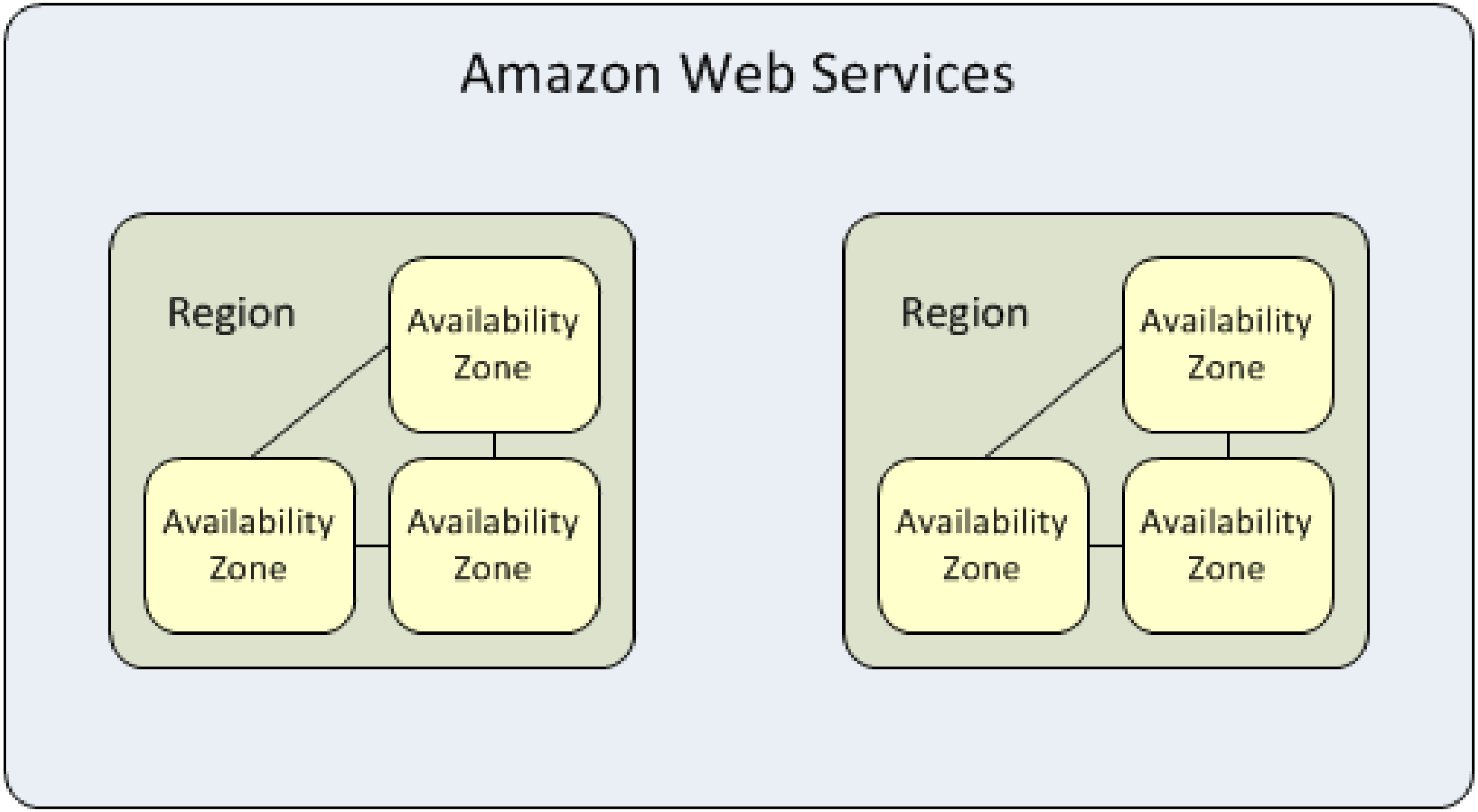

James (aws)
⌥⌘2
~ $ aws dynamodb wizard new-table
Enter the name of the table: MyTable
Enter the name of the primary key: pk
Primary key type
Add a sort key?
Select the read/write capacity mode.
Select Server-side encryption settings for your DynamoDB table
DEFAULT
KMS - AWS managed CMK
> KMS - Customer managed CMK



EC2




Instance
Host computer
AMI
Launch instances
of any type
Instances
Host computer


Amazon Web Services
Region
Availability Zone
Availability Zone
Availability Zone
Region
Availability Zone
Availability Zone
Availability Zone



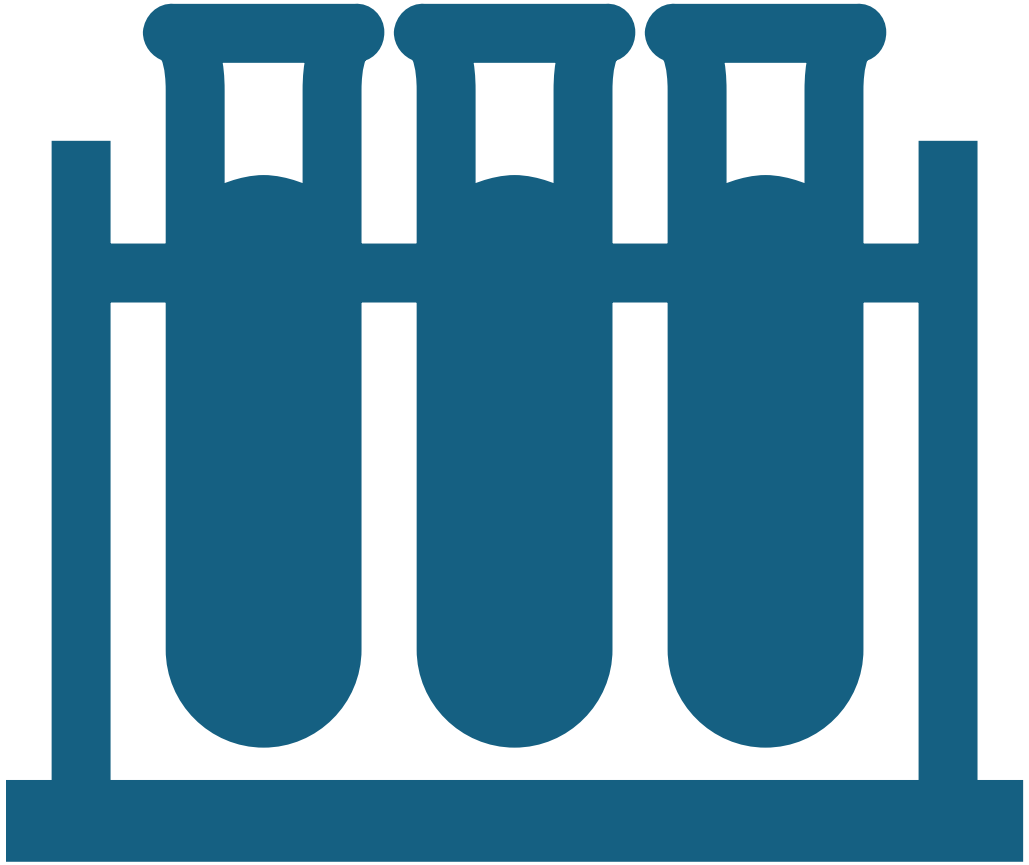
AWS LAMBDA

- Automatic provisioning and scaling: Lambda automatically handles the infrastructure needed to run your code, scaling up or down depending on the number of events or requests.
- Code monitoring and logging: It manages the monitoring of your code execution and logs activities for you, integrating with AWS CloudWatch.
- Pay only for what you use: You are not charged when your code is not running. Billing is based solely on the actual compute time your code consumes, measured in milliseconds.



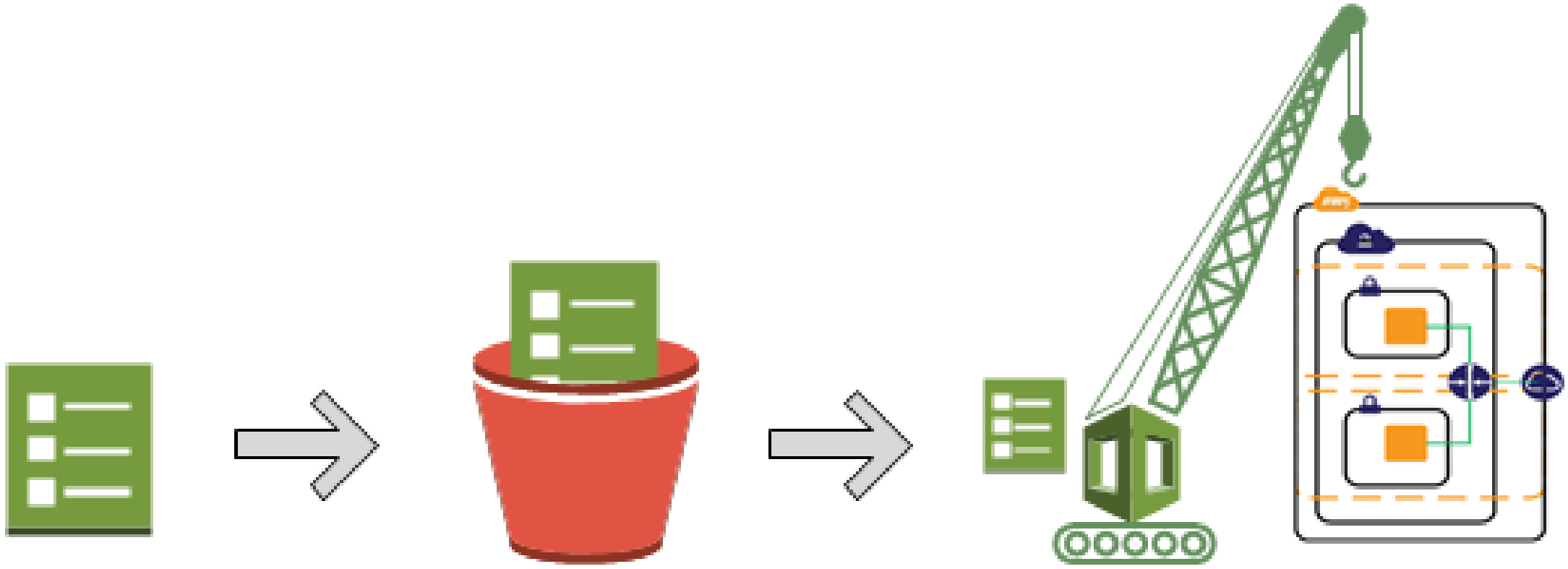

Photograph is taken
Amazon S3
Photo is uploaded to an S3 Bucket
Lambda is triggered
AWS Lambda
Lambda runs image resizing code
Photo is resized into web, mobile, and tablet sizes

EXERCISE 4





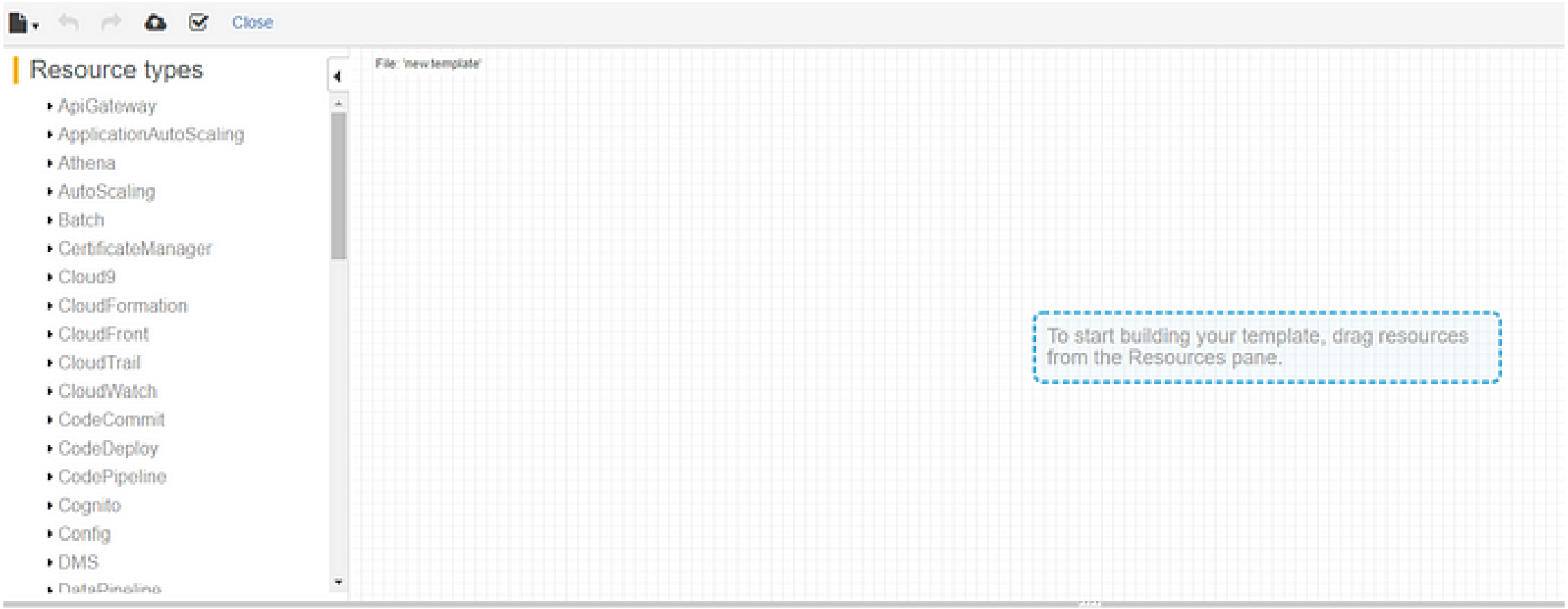
CLOUDFORMATION

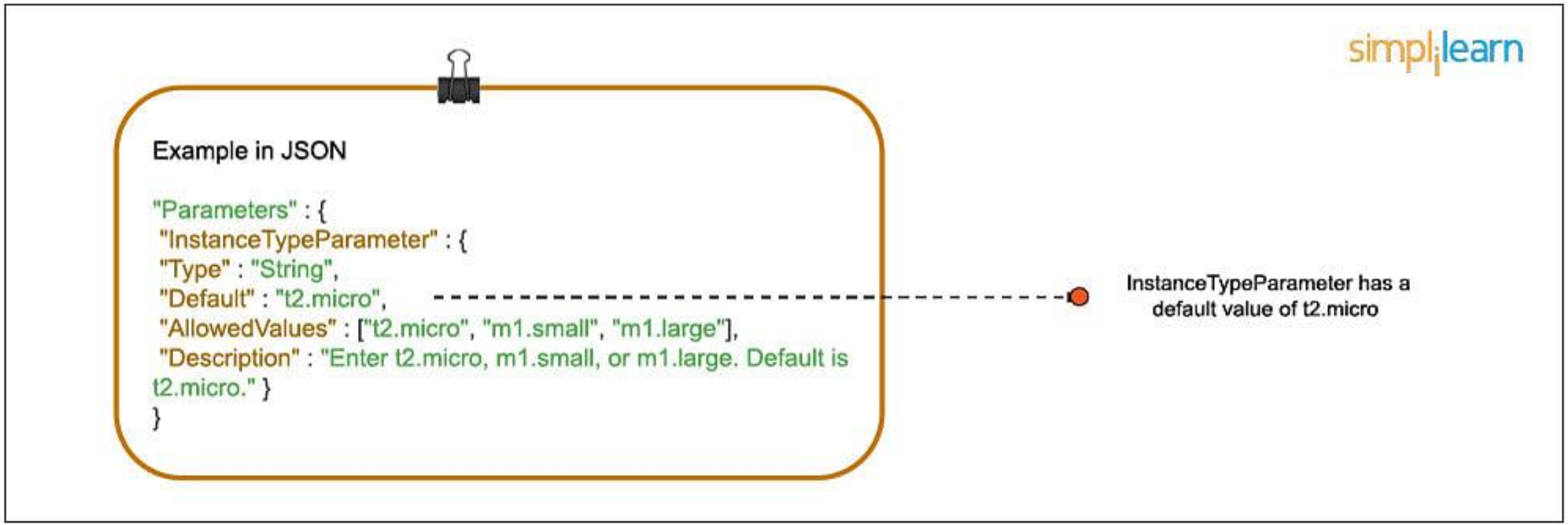


1 Create or use an existing template

2 Save locally or in S3 bucket

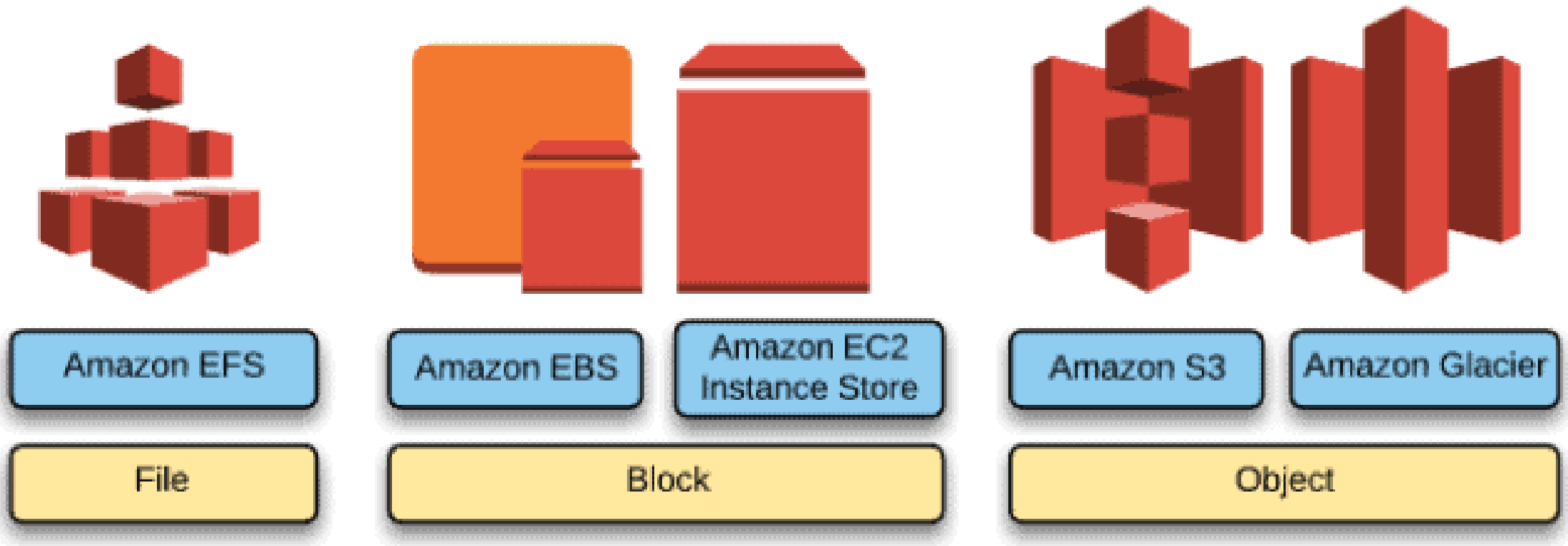
3 Use AWS CloudFormation to create a stack based on your template. It constructs and configures your stack resources.

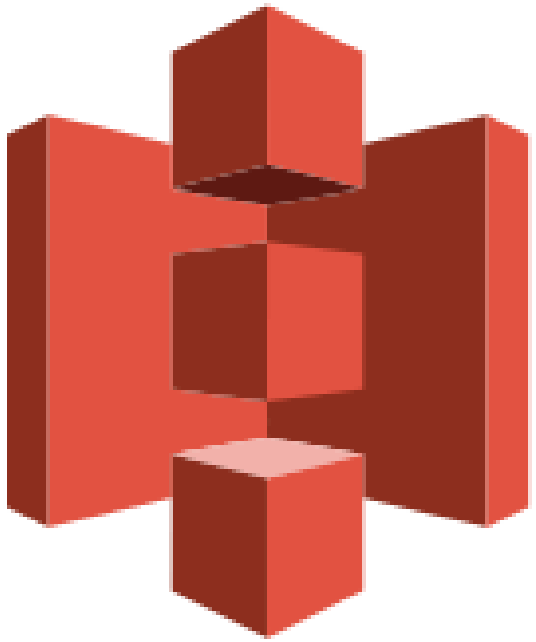

Close
Resource types
ApiGateway
ApplicationAutoScaling
Athena
AutoScaling
Batch
CertificateManager
Cloud9
CloudFormation
CloudFront
CloudTrail
CloudWatch
CodeCommit
CodeDeploy
CodePipeline
Cognito
Config
DMS
File: 'new.template'
To start building your template, drag resources from the Resources pane.
new.template
Choose template language:
{
"AWSTemplateFormatVersion": "2010-09-09"
}


simplilearn
Example in JSON
"Parameters" : {
"InstanceTypeParameter" : {
"Type" : "String",
"Default" : "t2.micro",
"AllowedValues" : ["t2.micro", "m1.small", "m1.large"],
"Description" : "Enter t2.micro, m1.small, or m1.large. Default is t2.micro." }
}
InstanceTypeParameter has a default value of t2.micro

EXERCISE 5




Amazon EFS
Amazon EBS
Amazon EC2
Instance Store
Amazon S3
Amazon Glacier
File
Block
Object



amazon
S3



S3 Standard

- Active, frequently accessed data
- Milliseconds access
- ≥ 3 AZ
- $0.0210/GB



S3 Intelligent-Tiering

- Data with changing access patterns
- Milliseconds access
- ≥ 3 AZ
- $0.0210 to $0.0125/GB
- Monitoring fee per object
- Min storage duration



S3 Standard-IA

- Infrequently accessed data
- Milliseconds access
- ≥ 3 AZ
- $0.0125/GB
- Retrieval fee per GB
- Min storage duration
- Min object size



S3 One Zone-IA

- Re-creatable, less accessed data
- Milliseconds access
- 1 AZ
- $0.0100/GB
- Retrieval fee per GB
- Min storage duration
- Min object size



S3 Glacier

- Archive data
- Select minutes or hours
- ≥ 3 AZ
- $0.0040/GB
- Retrieval fee per GB
- Min storage duration

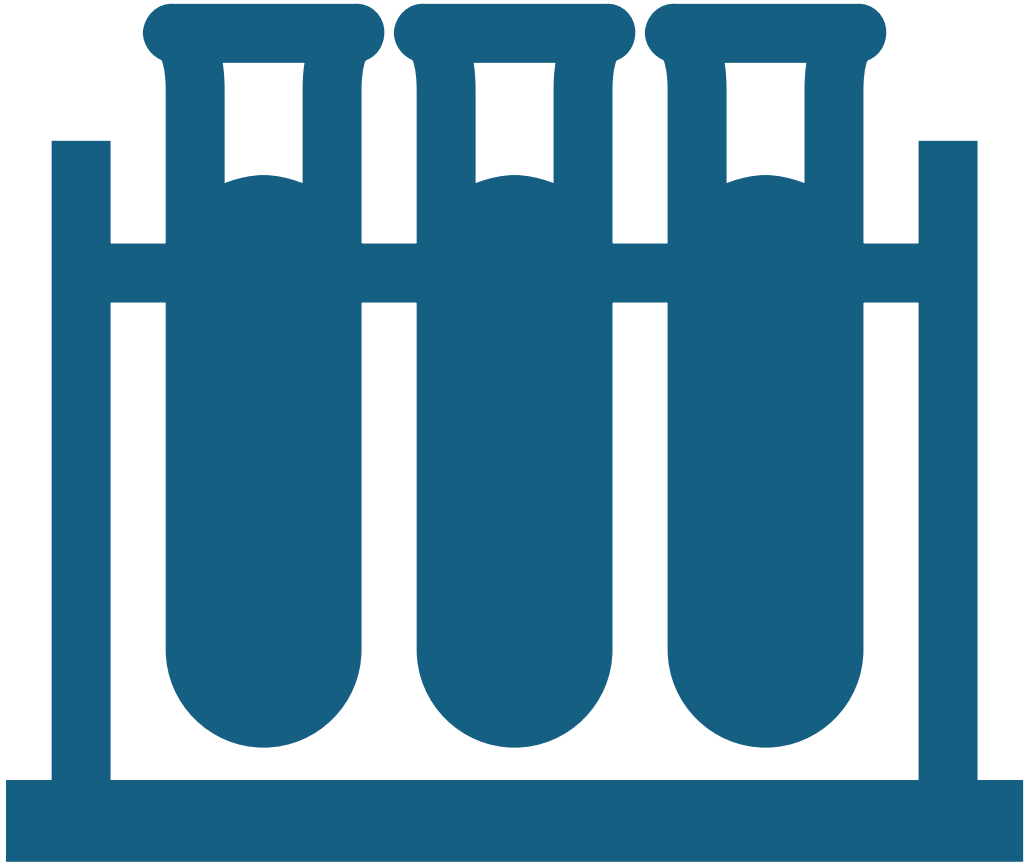


S3 Glacier Deep Archive

- Long-term archive-data
- Select hours
- ≥ 3 AZ
- $0.00099/GB
- Retrieval fee per GB
- Min storage duration

Frequent ***Access frequency*** ***Archive***

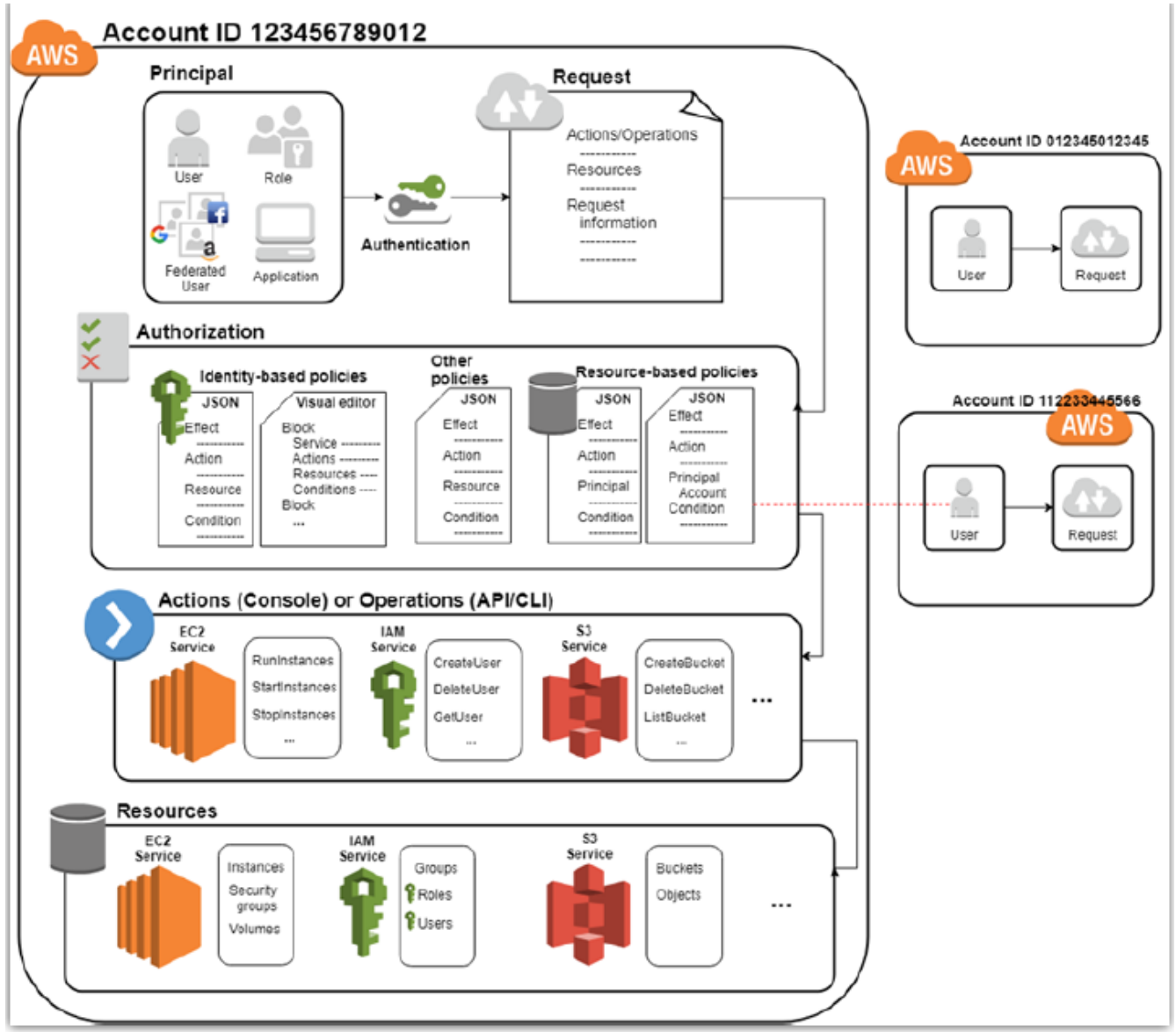
EXERCISE 6



IAM

- **Users: A user is a person who needs access to your AWS services. You can grant permissions to a user to control what they can and cannot access.**
- **Groups: A group is a collection of users who share the same permissions. This is often used to simplify management.**
- **Roles: A role is used to define a set of permissions and who can use them. However, it is not directly assigned to a person or service. Instead, a service or person can assume the role when needed. Roles are temporary and therefore provide greater security compared to granting permanent permissions through groups or users.**




Account ID 123456789012
AWS
Principal
User
Role
Federated User
Application
Authentication
Request
Actions/Operations
Resources
Request information
Authorization
Identity-based policies
JSON
Effect
Action
Resource
Condition
Visual editor
Block
Service
Actions
Resources
Conditions
Block
Other policies
JSON
Effect
Action
Resource
Condition
Resource-based policies
JSON
Effect
Action
Principal
Condition
JSON
Effect
Action
Principal
Account
Condition
Actions (Console) or Operations (API/CLI)
EC2 Service
RunInstances
StartInstances
StopInstances
IAM Service
CreateUser
DeleteUser
GetUser
S3 Service
CreateBucket
DeleteBucket
ListBucket
Resources
EC2 Service
Instances
Security groups
Volumes
IAM Service
Groups
Roles
Users
S3 Service
Buckets
Objects
Account ID 012345012345
AWS
User
Request
Account ID 112233445566
AWS
User
Request

Generated policy

1 2 3

Customize permissions

Review the following policy template. You must specify resources for actions that support resource-level permissions to continue creating the policy.

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "access-analyzer:ValidatePolicy",
                "iam:GetAccountPasswordPolicy",
                "iam:GetAccountSummary",
                "iam:ListAccountAliases",
                "iam:ListGroups",
                "iam:ListPolicies",
                "iam:ListRoles",
                "iam:ListUsers"
            ],
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "iam:GetRole",
                "iam:ListAttachedRolePolicies",
                "iam:ListInstanceProfilesForRole",
                "iam:ListRolePolicies",
                "iam:ListRoleTags"
            ],
            "Resource": "arn:aws:iam::${Account}:role/${RoleNameWithPath}"
        },
        {
            "Effect": "Allow",
            "Action": [
                "iam:GetUser",
                "iam:ListAccessKeys",
                "iam:ListAttachedUserPolicies",
                "iam:ListGroupsForUser",
                "iam:ListUserTags"
            ],
            "Resource": "arn:aws:iam::${Account}:user/${UserNameWithPath}"
```

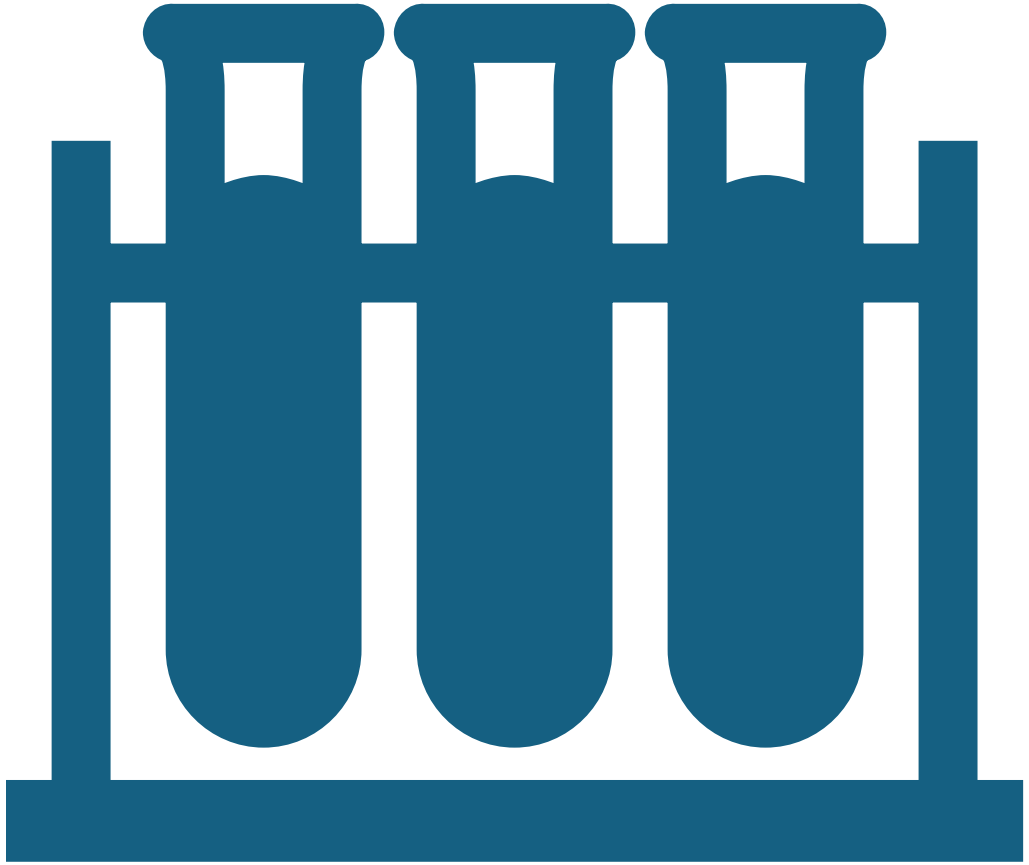
Edit statement

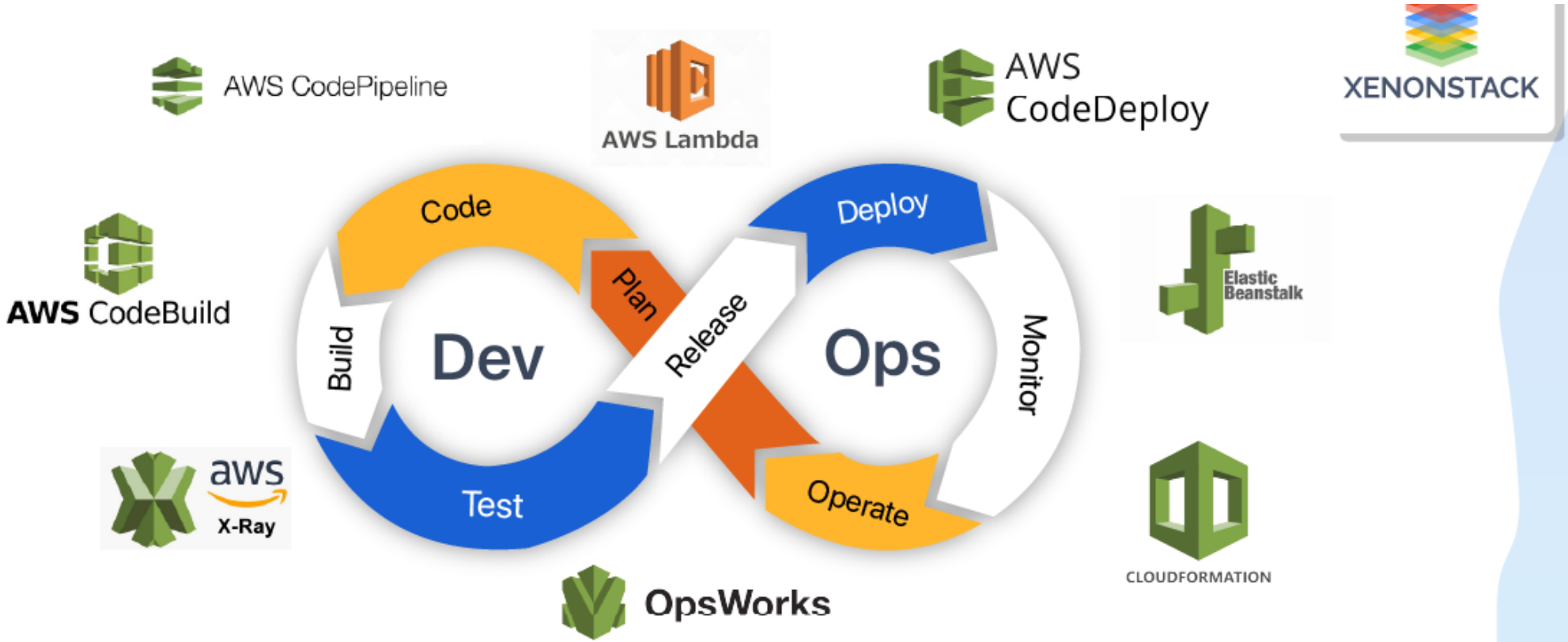
Select a statement

Select an existing statement in the policy or add a new statement.

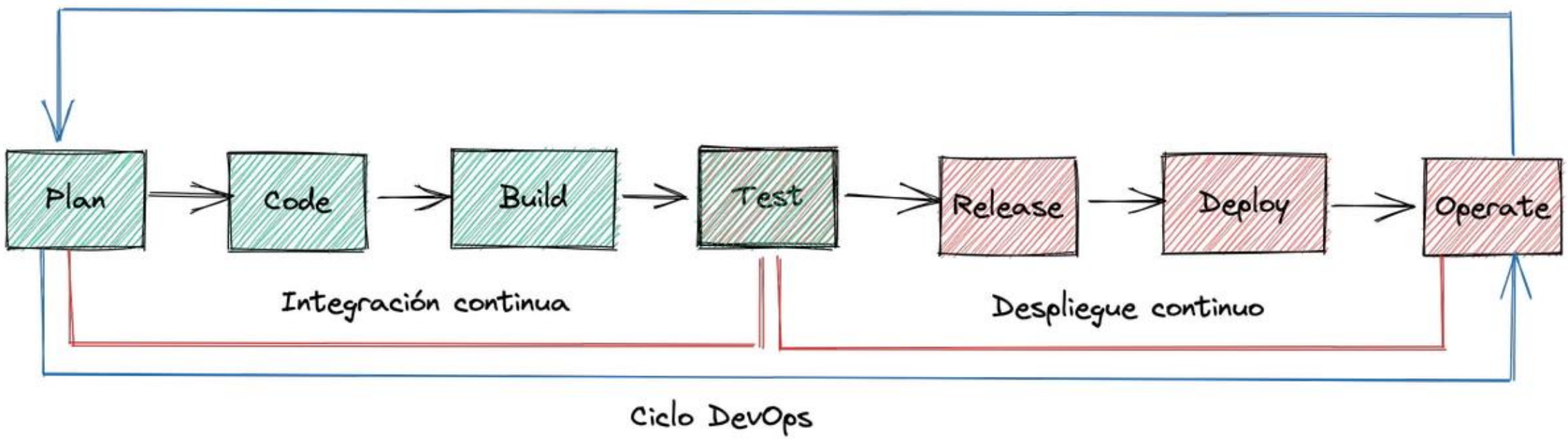
+ Add new statement

EXERCISE 7

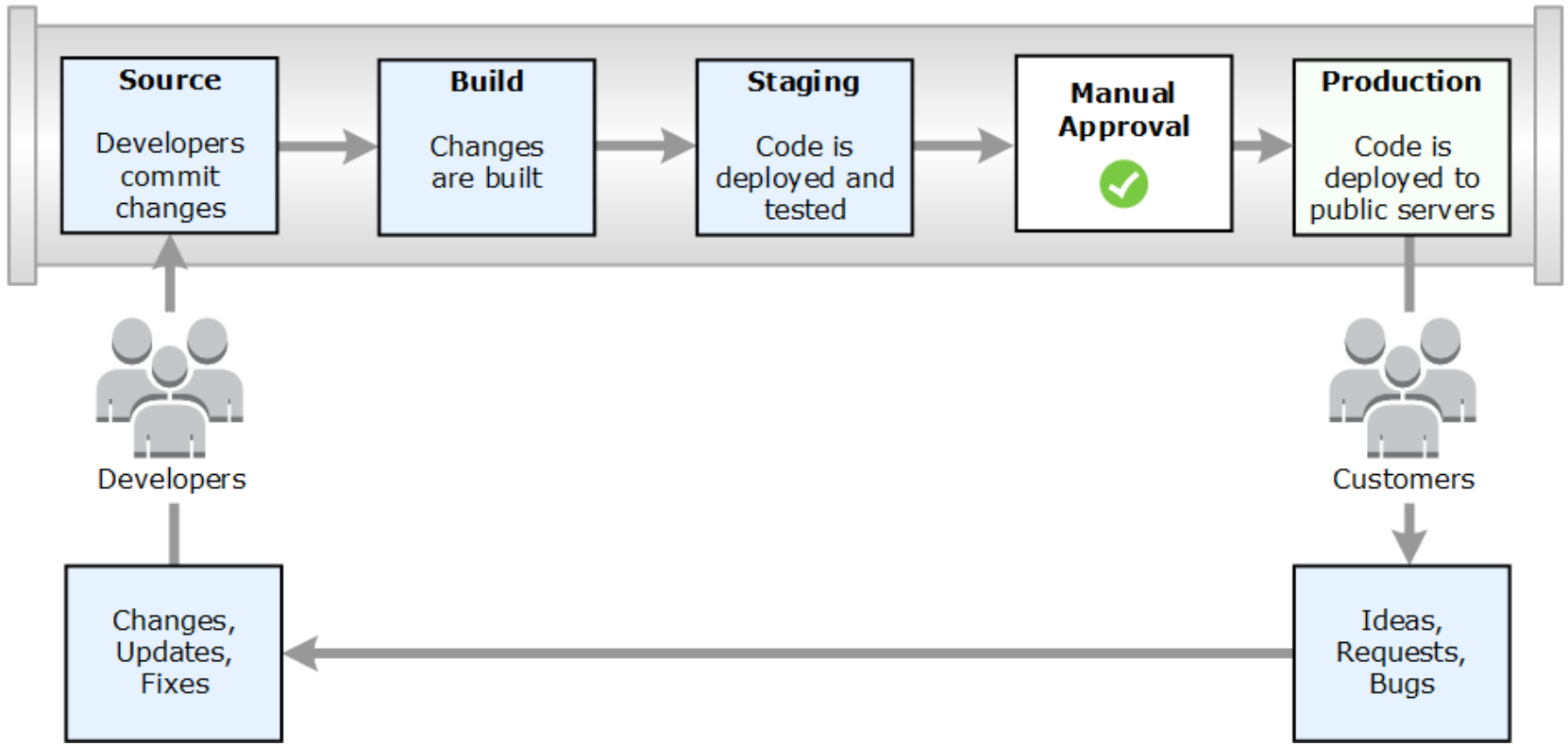


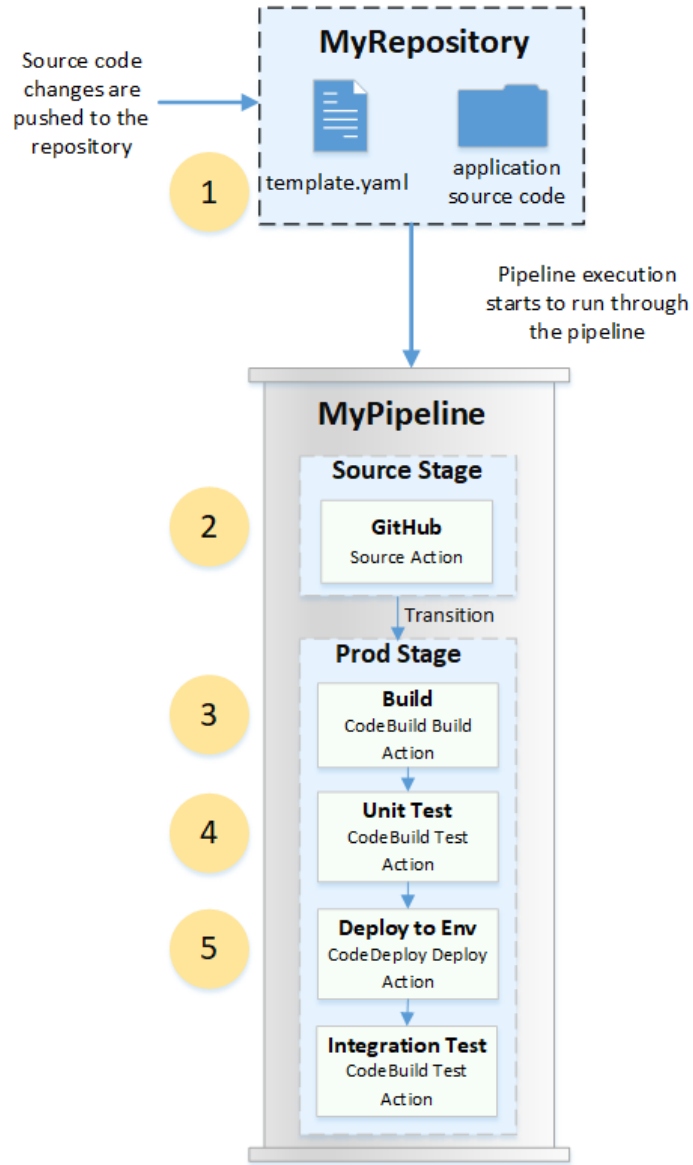


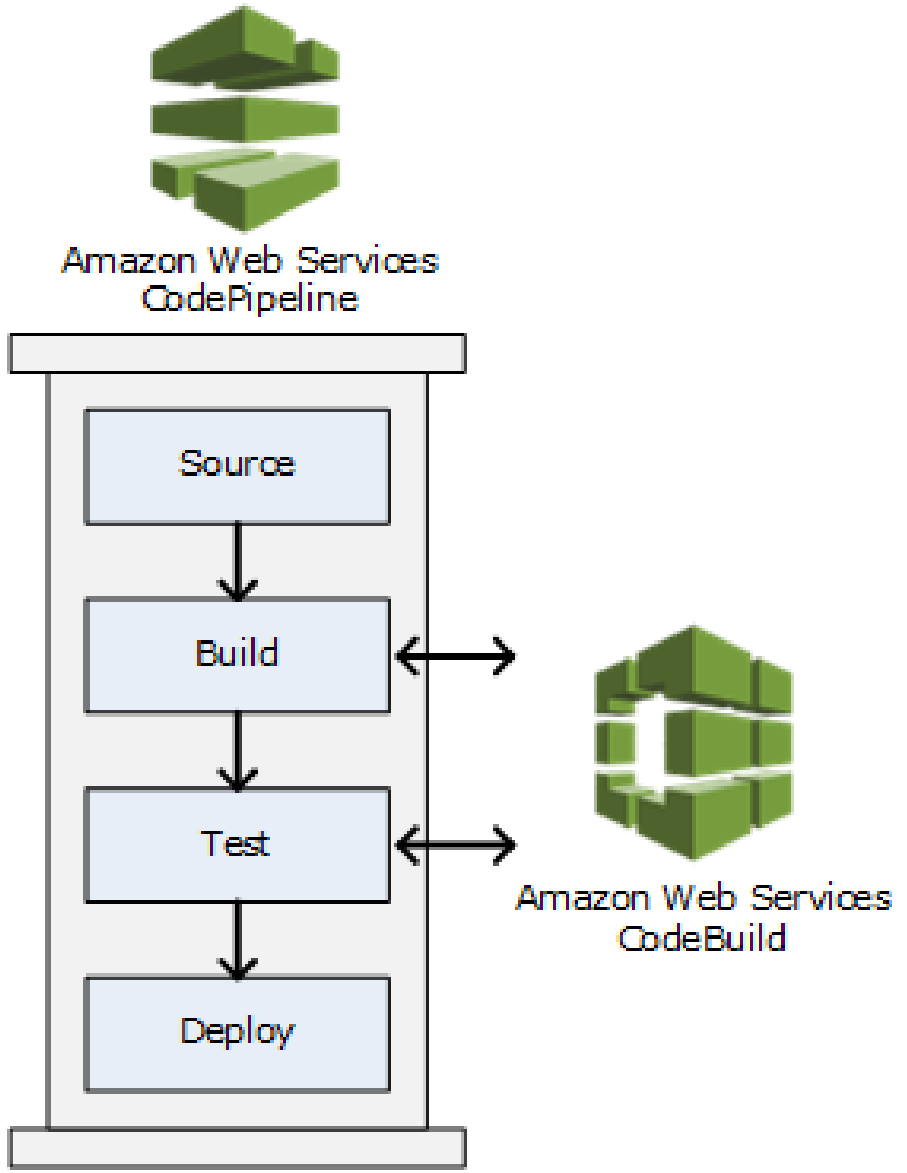
Top 8 AWS DevOps Tools and Use Cases



Ciclo DevOps


Source
Developers commit changes
Build
Changes are built
Staging
Code is deployed and tested
Manual Approval
Production
Code is deployed to public servers
Developers
Customers
Changes, Updates, Fixes
Ideas, Requests, Bugs


Source code changes are pushed to the repository
MyRepository
template.yaml
application source code
1
Pipeline execution starts to run through the pipeline
MyPipeline
Source Stage
2
GitHub
Source Action
Transition
Prod Stage
3
Build
CodeBuild Build Action
4
Unit Test
CodeBuild Test Action
5
Deploy to Env
CodeDeploy Deploy Action
Integration Test
CodeBuild Test Action

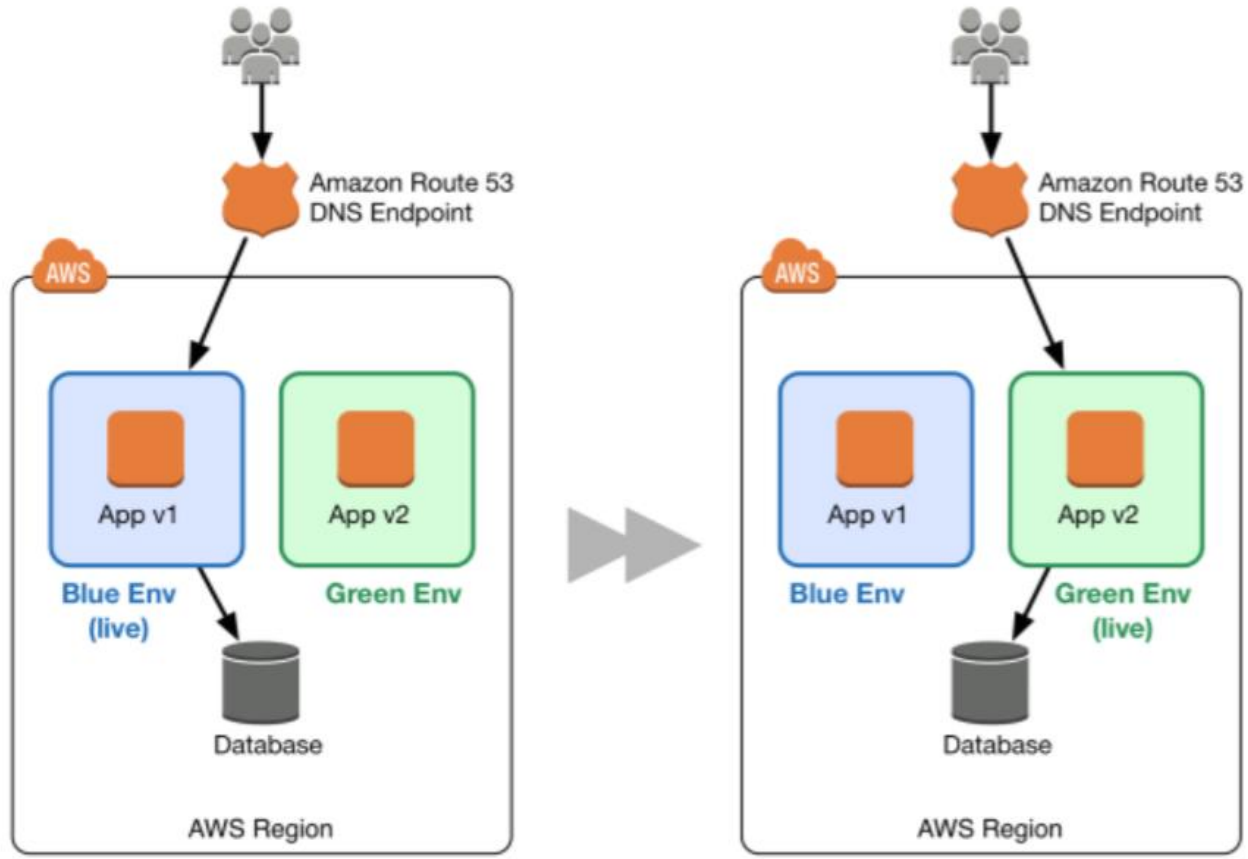

Amazon Web Services
CodePipeline
Source
Build
Test
Deploy
Amazon Web Services
CodeBuild

DEMO

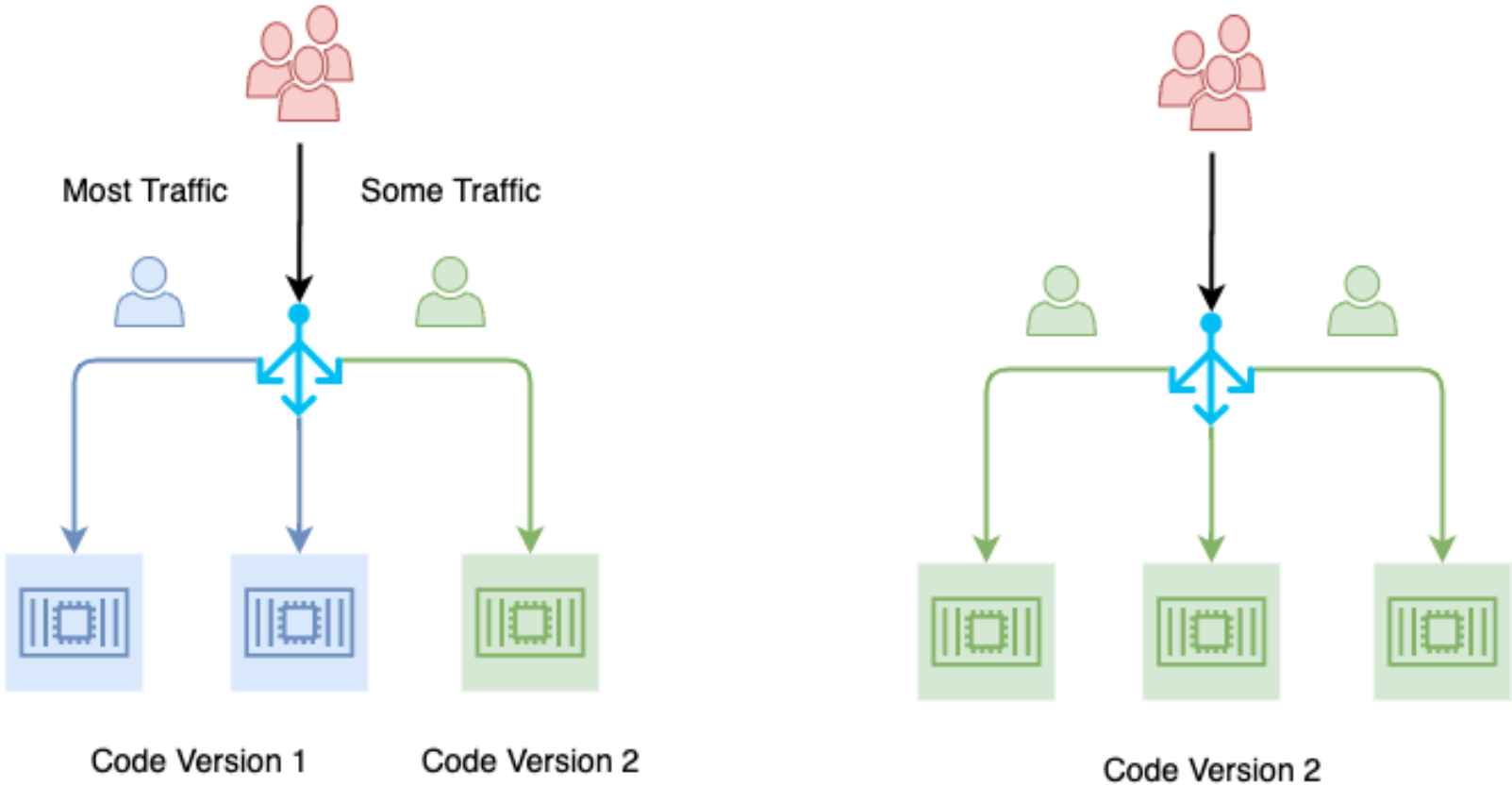


EXERCISE 8



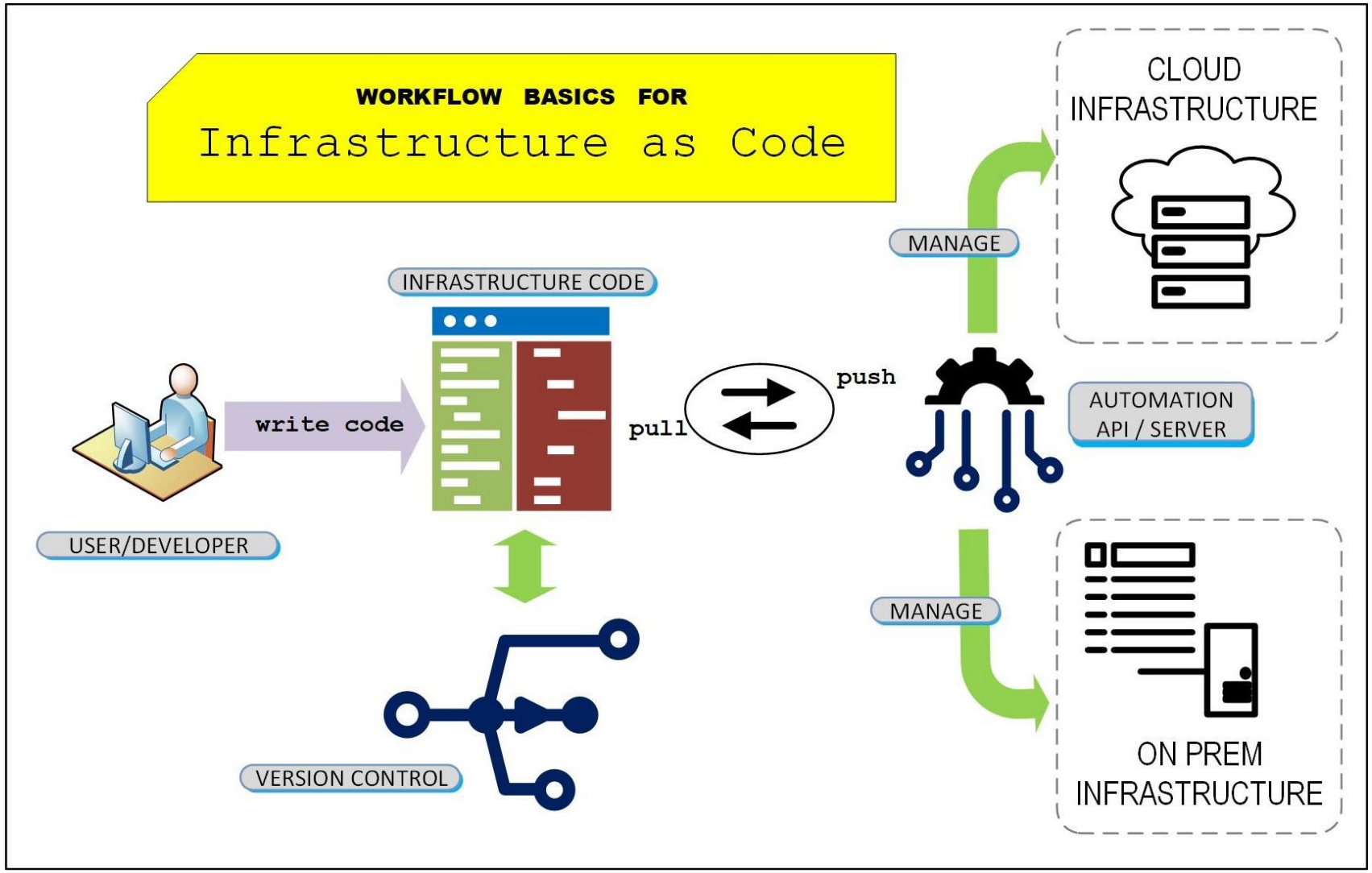

Amazon Route 53
DNS Endpoint
AWS
App v1
App v2
Blue Env
(live)
Green Env
Database
AWS Region
Amazon Route 53
DNS Endpoint
AWS
App v1
App v2
Blue Env
Green Env
(live)
Database
AWS Region

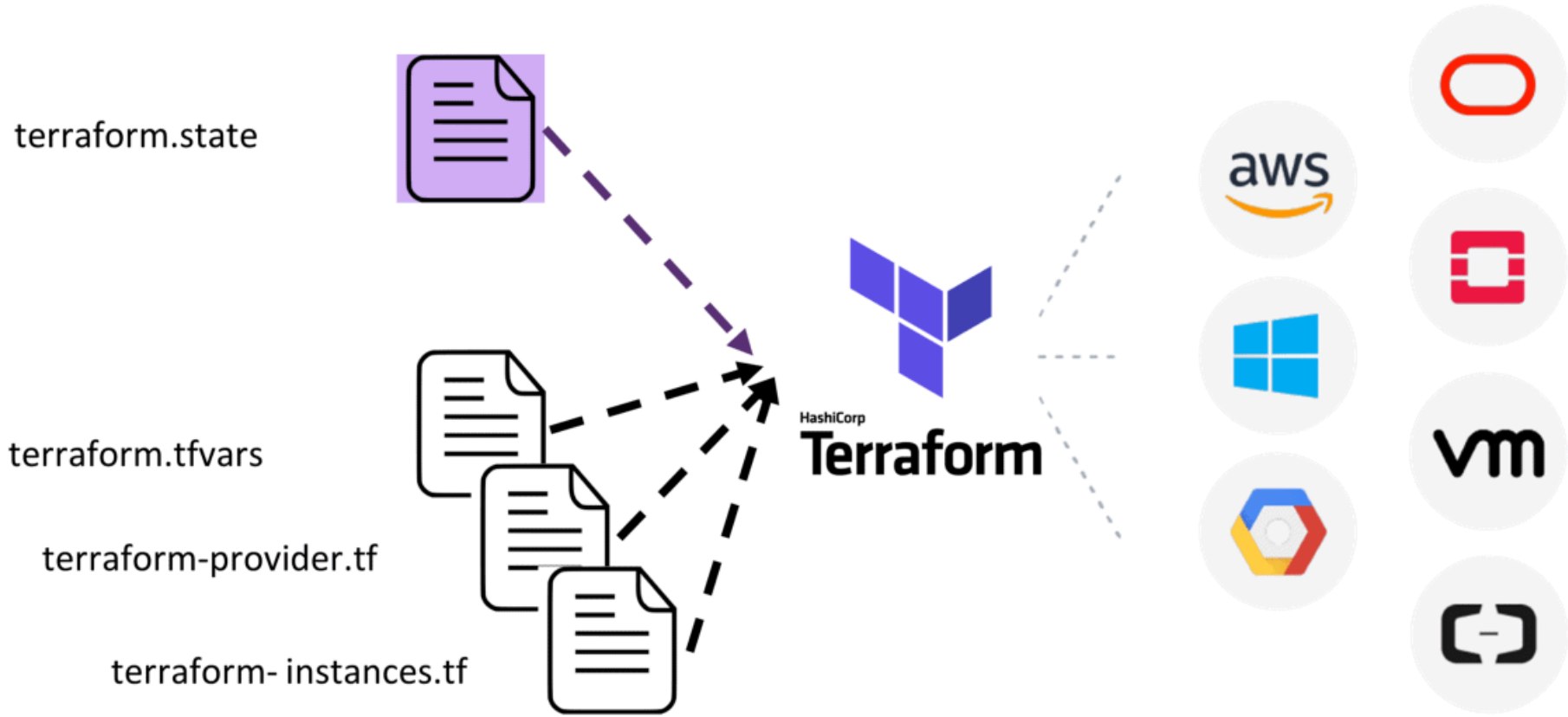
Canary Deployment



EXERCISE 9




WORKFLOW BASICS FOR
Infrastructure as Code
CLOUD INFRASTRUCTURE
MANAGE
INFRASTRUCTURE CODE
write code
pull
push
AUTOMATION API / SERVER
USER/DEVELOPER
MANAGE
VERSION CONTROL
ON PREM INFRASTRUCTURE


terraform.state
terraform.tfvars
terraform-provider.tf
terraform- instances.tf
HashiCorp
Terraform
aws

DEMO



EXERCISE 10

